

Foundational Data Analysis Skills

Chapter Preview

You will be working with data and data analysis software throughout your career, and one of the most common software tools used in accounting is Microsoft Excel. As you will see in the Professional Insight feature, the ability to use Microsoft Excel to manipulate large data sets is a tremendous asset for newly hired professional accountants. While Microsoft Excel skills are important, the software is not powerful enough to analyze extremely large data sets, so it is not the only tool used to analyze data. This chapter introduces some skills that, regardless of the technology used, are fundamental to performing data analytics. Microsoft Excel is used to demonstrate many of the core data analysis skills, but subsequent chapters also introduce data analysis software such as Power BI and Tableau. A combination of a core understanding of data, data visualization, and descriptive analysis skills are the foundation for performing more advanced data analytics.

Professional Insight: How Can Pivot Tables Help Make Sense of Large Data Sets?

Josh, a senior accounting student, explains how learning Microsoft Excel helped him in his internship.

My first assignment was to create an Excel PivotTable to manipulate a huge data file of approximately 450,000 records. I had just obtained my Microsoft Office Specialist Basic Excel certification when I began my internship at PwC. It feels great to apply what I learned in the classroom in the business world. **This experience gave me confidence interacting with my superiors and helped me build a reputation of reliability at my firm.** My director on the client was thoroughly impressed with my work, and my senior associate was surprised that I was able to put together such a comprehensive table with so little experience.

Chapter Roadmap

LEARNING OBJECTIVES	TOPICS	APPLY IT
LO 2.1 Describe how data is stored in and extracted from relational databases.	- Relational Databases - Joining Tables	Identify Primary and Foreign Keys (Example: Accounting Information Systems)
LO 2.2 Explain how functions help answer data analysis questions.	- Basic Functions for Data Analysis - Applying Excel Basic Functions	Analyze Sales Transactions with Excel Functions (Example: Financial and Managerial Accounting)
LO 2.3 Illustrate how pivot tables organize and filter data.	- Using Pivot Tables - Filtering Pivot Tables	Analyze Sales with Excel PivotTables (Example: Financial and Managerial Accounting)
LO 2.4 Identify descriptive measures used to perform data analysis.	- Measures of Location - Measures of Dispersion - Measures of Shape - Correlation Analysis	Use Descriptive Statistics to Audit Warranty Expense (Example: Auditing)
LO 2.5 Summarize how data visualization explores and explains data.	- Making Sense of Large Data Sets - Visualizations and When to Use Them - Microsoft Excel Visualizations	Analyze Product Costs with Data Visualization (Example: Managerial Accounting)

Data The Data tag appears in the chapter when the data for an example, illustration, or application are available on Wiley's online learning platform.

Data analytics software is continuously changing, and there may be more recent versions of the software referenced in this chapter. For more information access the accompanying video on Wiley's online learning platform.

2.1 How Does Understanding Data Storage Help Answer Questions?

LEARNING OBJECTIVE 1

Describe how data is stored in and extracted from relational databases.

Understanding how data are stored is critical for data analysis. This is because the kind of analysis that can be performed depends on the data being used, and identifying and extracting the data we need requires knowing how it is stored.

Relational Databases

Data, regardless of type or format, needs to be stored somewhere. One way to do that is in a **relational database**, which is a collection of logically related data that can be retrieved, manipulated, and updated to meet users’ needs. Most data you will work with in your accounting career will come from relational databases, where data are stored in individual tables that can be linked together. Once the tables are linked, data from multiple tables can be accessed. A **table** in a relational database stores valuable data related to an object of interest, such as a business resource, event, or agent. Tables are comprised of rows and columns:

- Each row represents one record or instance of the table’s object.
- The columns reflect the **attributes**, which are the data fields that describe different aspects of the records (**Illustration 2.1**).

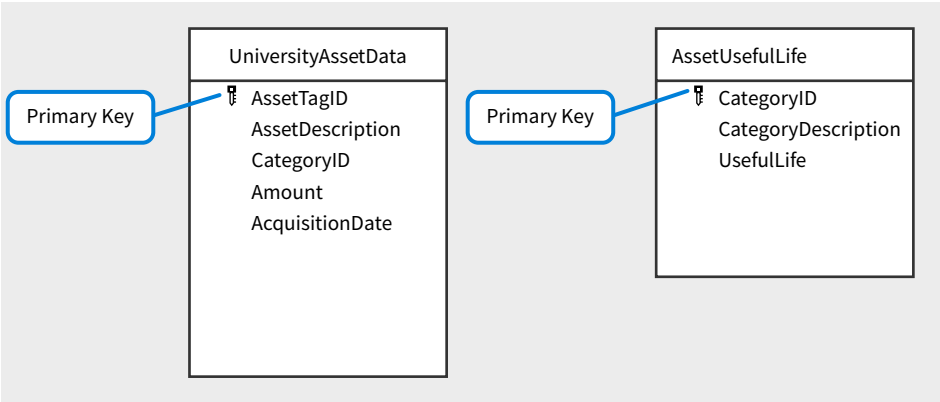
ILLUSTRATION 2.1 Database Elements and Examples of Tables and Attributes

Database Elements	Table Examples	Examples of Attributes
Resources: Identifiable objects that have economic value to the business entity.	Inventory items	• Inventory item’s code number • Description • Cost • Quantity on hand
Events: An organization’s business activities.	Sales orders	• Sales order number • The date of a sales order • Customer ID of the purchaser
Agents: Represent people or organizations about which data are collected.	Employees	• Employee number • Name • Address • Phone number



Illustration 2.2 is the database view of a university’s asset data table that contains data for the inventory of its assets.

ILLUSTRATION 2.2 University Database



Each row in the table shows a uniquely identifiable asset. Its associated attributes are organized in each vertical column listed below the table name:

- AssetTagID: Tag identification number
- AssetDescription: Description of asset
- CategoryID: Category identification number
- Amount: Cost of asset
- AcquisitionDate: Date asset was acquired

The AssetUsefulLife table contains information related to the asset categories. Each row is a unique category, and the columns are the attributes of that category:

- CategoryID: Identification number for each category of assets
- CategoryDescription: Category description
- UsefulLife: Useful life in years of assets in each category

There is a symbol for a key next to the AssetTagID attribute in the first table and the CategoryID in the second table. This symbol identifies the **primary key**, which is the column that must have a unique value for each row in the table. In the UniversityAssetData table, every asset is uniquely identified by the primary key column AssetTagID. In the AssetUsefulLife table, the primary key is CategoryID. Each row in the AssetUsefulLife table will have a unique CategoryID number.

To calculate depreciation on assets, we need cost information, useful life information, and the age of the asset. However, that information appears in two different tables. Linking them requires a common field in both:

- The CategoryID field is in both tables.
- CategoryID is the primary key column in the AssetUsefulLife table.
- In the UniversityAssetData table, CategoryID is a **foreign key** column. A foreign key column contains the same data as a primary key from another table. It has been repeated in the second table so the tables can be linked in relationship to each other. **Illustration 2.3** shows the relationship linking the two tables using the CategoryID attribute.

Why does this matter? Linking tables creates a relationship that makes it possible to pull information from both tables and create calculations of depreciation for every asset in the asset data table.

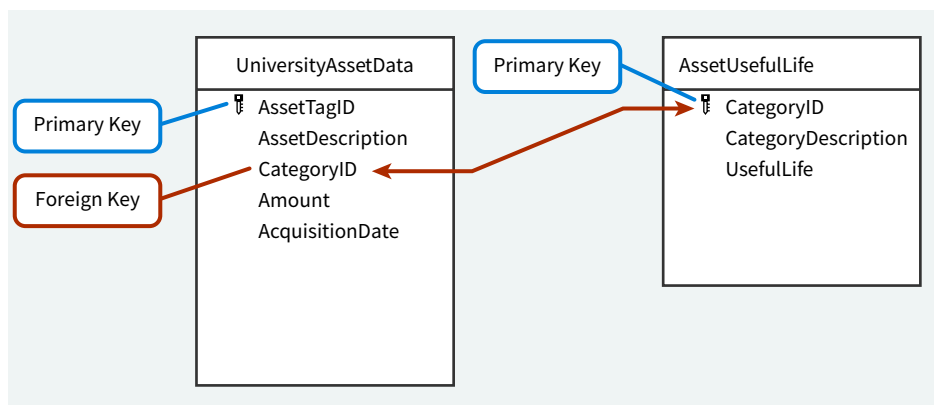


ILLUSTRATION 2.3 Creating a Relationship Between Tables

Joining Tables

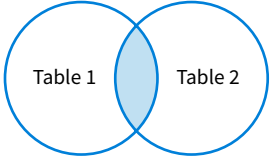
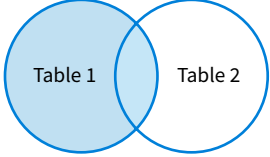
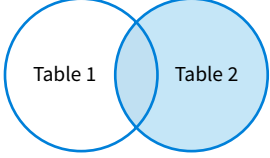
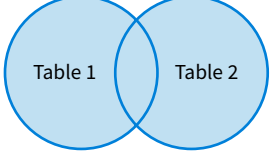
Once common fields using primary and foreign keys are identified across tables, the next step is to tell the database how to link the tables to extract data:

- A **query** is an action request made to a database. It provides the computer instructions for joining, adding, updating, deleting, retrieving, or manipulating the data in its tables. Queries can be created and used once or stored for later reuse.
- The standard query command language used for database management is **Structured Query Language (SQL)**. Writing SQL code is beyond the scope of this chapter, but fortunately, many software programs have built-in applications that automatically create the SQL code needed to query the database.

Retrieving all the data fields necessary to complete a particular task requires understanding how to join tables based on columns they have in common. Tables are linked by creating a **join** that combines rows from two or more tables based on a related column between them. Joins are also used in visualization software and other data analysis software when there is more than one table of data being used in an analysis. (We discuss data visualization in more detail both at the end of this chapter and in later chapters.)

The most common joins are inner, left, right, and full joins. **Illustration 2.4** summarizes the different joins and provides a visual representation. The blue shading represents the results of the join. Note that a null value is not the same as a value of zero. A **null value** is when a value is unknown or missing.

ILLUSTRATION 2.4 Types of Joins

Join	Description	Visual Representation
Inner	• Selects all rows from both tables with matching values. • The results will not have null values in any key column.	
Left	• Returns all records from the left table and the matching records from the right table. • There could be null values if there is no matching value in the right table to a left table record.	
Right	• Returns all records from the right table and the matched records from the left table. • Could return null values if there is not a matching value in the left table to the record in the right table.	
Full	• Returns all records when there is a match in either the left or right table. • Nulls could be present in this result if there are no matching records between the tables.	

We can illustrate how these joins work with an example. Bikes R Us is a bicycle wholesale dealer. **Illustration 2.5** shows two tables from their database:

- The left table is the Customer table, and the Order table is on the right.
- CustomerID is the primary key in the Customer table.
- OrderID is the primary key in the Order table. CustomerID is a foreign key.

ILLUSTRATION 2.5 Bikes R Us Database Tables

Customer Table				Order Table			
CustomerID	CompanyName	ContactLastName	State	OrderID	OrderDate	Amount	CustomerID
1001	Cycle Nation	Creebo	NY	50012	2/1/2025	\$ 578.23	1001
1002	Spins	Geck	NC	50013	2/2/2025	\$ 982.99	1002
1003	Little Town Bike Stores	Blunsom	TX	50014	2/3/2025	\$ 1,563.32	1004
1004	Triathletes Store	Rings	CA	50015	2/4/2025	\$ 300.12	1001
				50016	2/5/2025	\$ 639.99	102

An inner join on these two tables will result in a table with all the data from the tables that match on the CustomerID field (**Illustration 2.6**).

ILLUSTRATION 2.6 Results from an Inner Join for Bikes R Us

Inner Join						
CustomerID	CompanyName	ContactLastName	State	OrderID	OrderDate	Amount
1001	Cycle Nation	Creebo	NY	50012	2/1/2025	\$ 578.23
1001	Cycle Nation	Creebo	NY	50015	2/4/2025	\$ 300.12
1002	Spins	Geck	NC	50013	2/2/2025	\$ 982.99
1004	Triathletes Store	Rings	CA	50014	2/3/2025	\$ 1,563.32

The inner join shows the customer and order information for each customer that has placed an order:

- These results reveal that the customer with CustomerID = 1003, Little Town Bike Stores, has not made any purchases.
- OrderID 50016 does not have a match in the customer table for CustomerID = 102, so that order is not reflected in the joined tables.

A left join returns all the rows from the left table and will show any matching data from the right table. If there are no matching rows in the right table, then the fields without matches will be null (**Illustration 2.7**).

ILLUSTRATION 2.7 Results from a Left Join for Bikes R Us

Left Join							
CustomerID	CompanyName	ContactLastName	State	OrderID	OrderDate	Amount	CustomerID
1001	Cycle Nation	Creebo	NY	50012	2/1/2025	\$ 578.23	1001
1001	Cycle Nation	Creebo	NY	50015	2/4/2025	\$ 300.12	1001
1002	Spins	Geck	NC	50013	2/2/2025	\$ 982.99	1002
1003	Little Town Bike Stores	Blunsom	TX	NULL	NULL	NULL	NULL
1004	Triathletes Store	Rings	CA	50014	2/3/2025	\$ 1,563.32	1004

The left join in Illustration 2.7 shows all the customers from the Customer table and matching information from the Order table:

- CustomerID = 1003 is listed in this join, but because there is not a matching order, the results for the order fields are null.
- OrderID = 50016 is not reflected in the joined table results. There is no CustomerID = 102 in the Customer table, so that record is not picked up in the join.

A right join returns all the rows from the right table and will show any matching data from the left table (**Illustration 2.8**).

ILLUSTRATION 2.8 Results from a Right Join for Bikes R Us

Right Join

CustomerID	CompanyName	ContactLastName	State	OrderID	OrderDate	Amount	CustomerID
1001	Cycle Nation	Creebo	NY	50012	2/1/2025	\$ 578.23	1001
1001	Cycle Nation	Creebo	NY	50015	2/4/2025	\$ 300.12	1001
1002	Spins	Geck	NC	50013	2/2/2025	\$ 982.99	1002
1004	Triathletes Store	Rings	CA	50014	2/3/2025	\$ 1,563.32	1004
NULL	NULL	NULL	NULL	50016	2/5/2025	\$ 639.99	102

The right join in Illustration 2.8 shows all orders from the Order table and any matching customers from the Customer table:

- There is not a matching customer record for CustomerID = 102, so the fields from the Customer table will have null values for that record.
- If there are no matching rows in the left table, then the fields will have null values.

A full join will return all the rows from both tables (**Illustration 2.9**).

ILLUSTRATION 2.9 Results from a Full Join for Bikes R Us

Full Join

CustomerID	CompanyName	ContactLastName	State	OrderID	OrderDate	Amount	CustomerID
1001	Cycle Nation	Creebo	NY	50012	2/1/2025	\$ 578.23	1001
1001	Cycle Nation	Creebo	NY	50015	2/4/2025	\$ 300.12	1001
1002	Spins	Geck	NC	50013	2/2/2025	\$ 982.99	1002
1003	Little Town Bike Stores	Blunsom	TX	NULL	NULL	NULL	NULL
1004	Triathletes Store	Rings	CA	50014	2/3/2025	\$ 1,563.32	1004
NULL	NULL	NULL	NULL	50016	2/5/2025	\$ 639.99	102

The full join shown in Illustration 2.9 shows all the records from both tables:

- Note the null values for the Order table fields for CustomerID = 1003 and the null values for the Customer table fields for OrderID = 50016.
- Any unmatched fields will have null values.

Joins are essential when analyzing data from multiple sources. **Illustration 2.10** shows examples of some questions accountants may need to ask of the data and the type of join that will answer them.

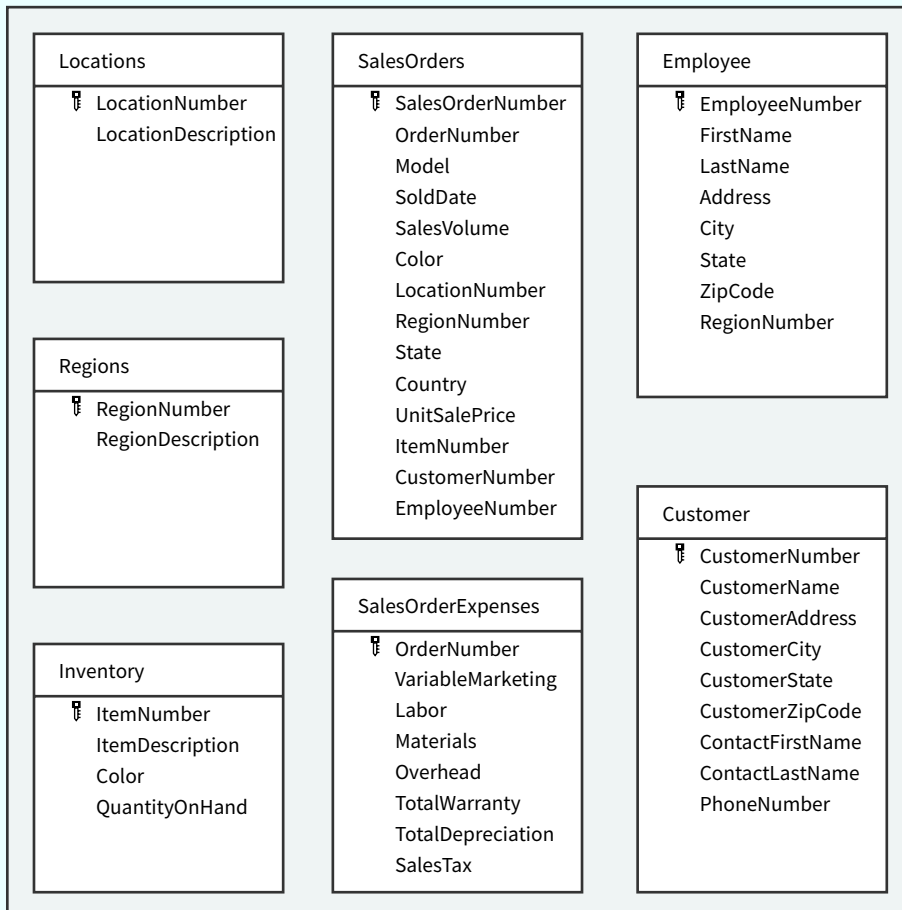
ILLUSTRATION 2.10 Sample Data Analysis Questions and Appropriate Join

Questions	Join
Are there any customer invoices with out receipts?	Left join
Have any transactions been charged to accounts that do not exist?	Left join
Have any payroll checks been issued to false employees?	Left join
Are there any sales returns to customers that do not exist?	Left join
Are there any inventory items that have not been sold? (Orders table on left)	Right join
Are there any customers that have not made a purchase? (Orders table on left)	Right join
Are there any vendors that have not been sent a purchase order? (PurchaseOrder table on left)	Right join
Are there any vendor invoices that have not been paid? (Payments table on left)	Right join
Do any employee addresses match vendor addresses?	Inner join
Do any employee phone numbers match vendor phone numbers?	Inner join
Have sales been made in a specific region?	Inner join

Accounting Information Systems Super Scooters, a manufacturer of motorized scooters, has just converted to a relational database system. You work in the accounting information systems area of the company and have been helping with the conversion to the new database system. The following tables have been established.

Apply It 2.1

Identify Primary and Foreign Keys



Your boss asks you to identify all primary and foreign keys for the tables in the database so joins can be created. For each table, list the primary key and any foreign keys.

SOLUTION

Table	Primary Key	Foreign Key
SalesOrders	SalesOrderNumber	OrderNumber LocationNumber RegionNumber CustomerNumber EmployeeNumber
Locations	LocationNumber	None
Region	RegionNumber	None
Inventory	ItemNumber	None
SalesOrderExpenses	OrderNumber	None
Employee	EmployeeNumber	RegionNumber
Customer	CustomerNumber	None

2.2 How Do Spreadsheet Functions Analyze Large Amounts of Data?

LEARNING OBJECTIVE 2

Explain how functions help answer data analysis questions.

Data analysis usually includes performing calculations such as adding amounts, counting data entries, and calculating averages. Frequently used calculations are often built into analysis software as **functions**, which are predefined formulas that perform calculations. One example is the SUM function in Microsoft Excel that adds a range of numbers in rows or columns.

Functions make it possible to quickly analyze large amounts of data without writing complex formulas. In fact, some of the most powerful attributes of Microsoft Excel are the built-in functions that perform calculations. Keep in mind that the most common functions and the logic behind them also apply to software other than Microsoft Excel. For example, functions can be used in analysis and visualization tools like Power BI and in Tableau. Understanding how these functions work, and more importantly, when to use them is a core data analysis skill.

Basic Functions for Data Analysis

Illustration 2.11 describes some basic Excel functions used in data analysis:

- The function name appears in the first column.
- The second column shows the function argument, which is the syntax necessary to invoke the function and the range and criteria to apply to it. All Excel functions begin with an equal sign, followed by the type of function being performed, and then parentheses that specify the arguments for the function. For example, to sum a column of numbers in column C and rows 2 through 245, the function would be: =SUM(C2:C245).
- The function calculation is described in the third column.

ILLUSTRATION 2.11 Basic Microsoft Excel Functions

Function	Function Arguments	Function Calculation
IF	=IF(logic test, value if true, value if false)	Returns one value if the condition is true and another one if it is false.
AVERAGE	=AVERAGE(Range)	Returns the arithmetic mean of the range, array, or numbers.
AVERAGEIF	=AVERAGEIF (Range, Criteria, AverageRange)	Finds the arithmetic mean for the cells specified by a given condition or criteria.
AVERAGEIFS	=AVERAGEIFS(SumRange, CriteriaRange1, Criteria1, CriteriaRange2, Criteria2)	Finds the arithmetic mean for the cells specified by a given set of conditions or criteria. Additional ranges, criteria ranges, and criteria can be added.
COUNT	=COUNT(Range)	Counts the number of cells in a range that contain numbers.
COUNTIF	=COUNTIF(Range, Criteria)	Counts the number of cells within a range that meet the given criteria.
COUNTIFS	=COUNTIFS(Range1, Criteria1, Range2, Criteria2)	Counts the number of cells specified by a given set of criteria. Additional ranges, criteria ranges, and criteria can be added.
COUNTA	=COUNTA(Range)	Counts the number of cells containing text in a range.
COUNTBLANK	=COUNTBLANK(Range)	Counts the number of blank cells in a range.
SUM	=SUM(Range)	Adds the cells in a range.
SUMIF	=SUMIF(Range, Criteria, SumRange)	Adds the cells specified by a specified condition or criteria.
SUMIFS	=SUMIFS(SumRange, CriteriaRange1, Criteria1, CriteriaRange2, Criteria2)	Adds the cells specified by a given set of conditions or criteria. Additional ranges, criteria ranges, and criteria can be added.

Returning to the university data example, **Illustration 2.12** shows the function arguments box for a COUNTIF function used to determine how many pumps the university owns. There are two input options to execute the functions illustrated:

- Type the function argument directly into a cell on the spreadsheet, or
- use the Function Arguments box.

ILLUSTRATION 2.12 University Asset Data COUNTIF Function Arguments Box

Function Arguments

COUNTIF

Range

B2:B13127

↑

= {"PUMP";"BONDER";"IRRIGATION SYSTEM";"DRI..."

Criteria

"Pump"

↑

= "Pump"

= 92

Counts the number of cells within a range that meet the given condition.

Range

is the range of cells from which you want to count nonblank cells.

Formula result = 92

[Help on this function](#)

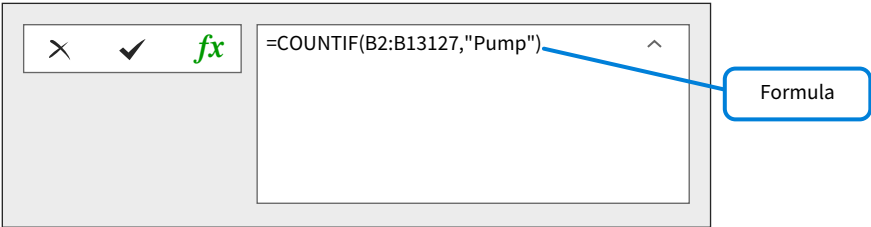
OK

Cancel

To open:

- Click on the **fx** next to the cell formula, which is above the spreadsheet (see the green **fx** in [Illustration 2.13](#)).
- Next, the **Function Arguments** input box in [Illustration 2.12](#) appears on the screen.
- Fill in the range and criteria, and select **OK**. The formula for the function then appears ([Illustration 2.13](#)). Note that the criteria (in this example “Pump”) must be typed in quotations. This is true for any criteria that is not a cell reference or a number.

ILLUSTRATION 2.13 Finding the Function Arguments Symbol



These Excel functions can help quickly analyze data sets, especially when they are very large. Let’s now apply these functions to answer questions from a data set about fixed assets.

Applying Excel Basic Functions

Data We will use the university’s assets data set ([Illustration 2.14](#)) to illustrate how functions can help make sense of data.

ILLUSTRATION 2.14 University Asset Data

University Asset Data						
	A	B	C	D	E	F
1	AssetTagID	AssetDescription	CategoryID	Category	Amount	AcquisitionDate
2	U000009	Pump	15	Laboratory Equipment	\$ 7,826.00	9/2/2021
3	U000010	Bonder	15	Laboratory Equipment	\$ 28,628.01	9/11/2021
4	U000015	Irrigation System	1	Agricultural and Farm Machinery	\$ 17,068.00	9/29/2020
5	U000016	Drill	1	Agricultural and Farm Machinery	\$ 28,344.00	10/27/2019
13126	U102879	Server	7	Computer Equipment	\$ 53,008.90	10/9/2021
13127	U204391	Oven	11	Food Preparation & Serving Equipment	\$ 7,255.53	5/6/2023

The data set was created from the University Asset Data file:

- Each row in the data set represents a unique asset owned by the university.
- The columns represent the attributes for each asset.

There are 13,127 rows of data in this spreadsheet, so visually scanning them is impossible. Instead, take advantage of the available Excel functions. Imagine you are responsible for reviewing the fixed assets for the university. You might want to ask the questions listed in [Illustration 2.15](#). Each question can be answered using an Excel function.

ILLUSTRATION 2.15 Questions, Functions, and Answers Using University Asset Data

Question	Function	Answer
What is the total cost of fixed assets?	=SUM(E2:E13127)	\$225,069,282.74
How many fixed assets does the university have in total?	=COUNTA(A2:A13127)	13,126
How many assets were acquired after 2022?	=COUNTIF(F2:F13127,">12/31/2022")	1,388
What is the total cost of assets acquired after 2022?	=SUMIF(F2:F13127,">12/31/2022",E2:E13127)	\$27,653,067.52
Are any descriptions missing?	=COUNTBLANK(D2:D13127)	0

Illustration 2.15 uses the SUM, COUNTA, COUNTIF, SUMIF, and COUNTBLANK functions. Could a SUMIFS or a COUNTIFS also be used?

If the question was how many computers the university purchased in 2023, then use the COUNTIFS function because there are two criteria. Recall that the COUNTIFS function requires specifying the range of cells to apply to the first criteria (computer equipment). The next criteria are purchases made in 2023, and the range will be the column with the dates acquired. [Illustration 2.16](#) shows the Function Arguments box for the COUNTIFS function and the result.

ILLUSTRATION 2.16 Function Arguments for COUNTIFS function

Function Arguments

COUNTIFS

Criteria_range1	D2:D13127	= {"Laboratory Equipment";"Labora
Criteria1	"Computer Equipment"	= "Computer Equipment"
Criteria_range2	F2:F13127	= {44441;44450;44103;43765;44446;4446
Criteria2	"> 12/31/2022"	= "> 12/31/2022"
Criteria_range3	F2:F13127	= {44441;44450;44103;43765;44446;4446
Criteria3	"< 1/1/2024"	= "< 1/1/2024"
Criteria_range4		= reference

= 283

Counts the number of cells specified by a given set of conditions or criteria.

Criteria_range4: is the range of cells you want evaluated for the particular condition.

Formula result = 283

[Help on this function](#)

OK

Cancel

The function box creates a formula:

=COUNTIFS(D2:D13127,"Computer Equipment",F2:F13127,">12/31/2022",
F2:F13127,"<1/1/2024")

To retrieve only the year 2023 data, create two criteria in the function argument:

- First, the date must be greater than December 31, 2022 (Criteria2).
- Second, the date must be less than January 1, 2024 (Criteria3).

By including those criteria in the argument, only results for the year 2023 are displayed.

Basic Excel functions are useful for answering questions with a single answer or an answer with one dimension. **Dimensions** are the variables or other fields that can be used to drill down or disaggregate the analysis measures. In other words, dimensions are used when there is a specific question about a specific aspect of the data.

For example, a question about how many computers were purchased in 2023 has only one dimension—computers in 2023. What about a question with more than one dimension? What if we want to know the sum of costs for each of the asset categories rather than just computers in a specific year? That question has more than one dimension because it involves every asset category and all years. We discuss how to address multi-dimension questions next.

Apply It 2.2

Analyze Sales Transactions with Excel Functions

Data **Financial Accounting** **Managerial Accounting** Super Scooters manufactures and sells four models of stand-on scooters: Celeritas, Captain, Lazer, and Kicks. Their customers range from large scooter-sharing companies to small retailers. Your supervisor has given you, an accountant for the company, a list of questions to answer using the Super Scooters data set. What follows is a portion of the Super Scooters Sales Transactions for the years 2023–2025. There are 3,645 transactions in the sales database.

	A	B	C	D	E	F	G	H	I	J
1	OrderNumber	Model	SoldDate	SalesVolume	Color	Location	State	DaysInInventory	UnitSalePrice	GrossSales
2	13684	Celeritas	12/31/2025	20	Yellow	Dallas	TX	83	\$ 342.00	\$ 6,840
3	13685	Lazer	12/31/2025	20	Blue	Seattle	WA	52	\$ 414.00	\$ 8,280
4	13682	Captain	12/30/2025	20	Green	Seattle	WA	29	\$ 679.00	\$ 13,580
5	13683	Lazer	12/30/2025	38	Green	Miami	FL	22	\$ 376.00	\$ 14,288
3645	10957	Lazer	12/22/2023	18	Yellow	Phoenix	AZ	17	\$ 330	\$ 5,940
3646	10960	Lazer	12/31/2023	35	Red	Phoenix	AZ	24	\$ 357	\$ 12,495
3647										



Explain which Microsoft Excel functions are best suited to answer the following questions about Super Scooters sales data.

1. What are the total gross sales dollars?
2. What are the total gross sales dollars for 2023?
3. How many sales transactions are there for the Lazer model?
4. What are the average gross sales for the Celeritas model in 2023?
5. How many green colored Captain models were sold?
6. If Super Scooters would like to add a column that indicates if the days on the lot are greater than 50 days, what function would be appropriate?

SOLUTION

1. SUM
2. SUMIF
3. CountIF
4. AverageIFS
5. CountIFS
6. IF

2.3 How Do We Organize Data Sets for Analysis?

LEARNING OBJECTIVE 3

Illustrate how pivot tables organize and filter data.

You have just learned how to use functions to answer questions with a single dimension. While you could use several functions to answer questions that involve multiple dimensions, it is more efficient to first use a data organization technique on the data set.

Data organization is the process of rearranging data to make it easier to understand. A **pivot table** is a tool that summarizes and reorganizes selected columns and rows of data in a spreadsheet, database, or business intelligence program. Pivot tables can quickly rearrange data to help answer many important business questions. Recall from the chapter opening professional insight that Josh needed to use one to reorganize a spreadsheet with 450,000 rows of data. It is unlikely Josh could have analyzed the spreadsheet data efficiently without a pivot table.

The examples here use Microsoft Excel to demonstrate how to create and filter a pivot-table. (Note that pivot tables created in this software are generally labeled as PivotTables.) Although the demonstrations use Excel PivotTables, these techniques are also used in other data analysis software. For example, they are useful when creating visualizations. Whichever data analysis tool you use, understanding the basic functionality of creating useful pivot tables and how to filter them is essential.

Using Pivot Tables

Both powerful and easy to use, a pivot table is also one of the most common tools you will use in your accounting career. It has five main components:

1. **Fields:** The data elements available for use in the pivot table.
2. **Columns:** When a field is chosen for a column area, only the unique values of the field are listed across the top.
3. **Rows:** When a field is chosen for the row area, it populates it as the first column. All row values are unique values and duplicates are removed.
4. **Values:** Each value is kept in a pivot table cell and displays the summarized information. Examples are sum, average, or count.
5. **Filters:** Apply a restriction to the entire table.

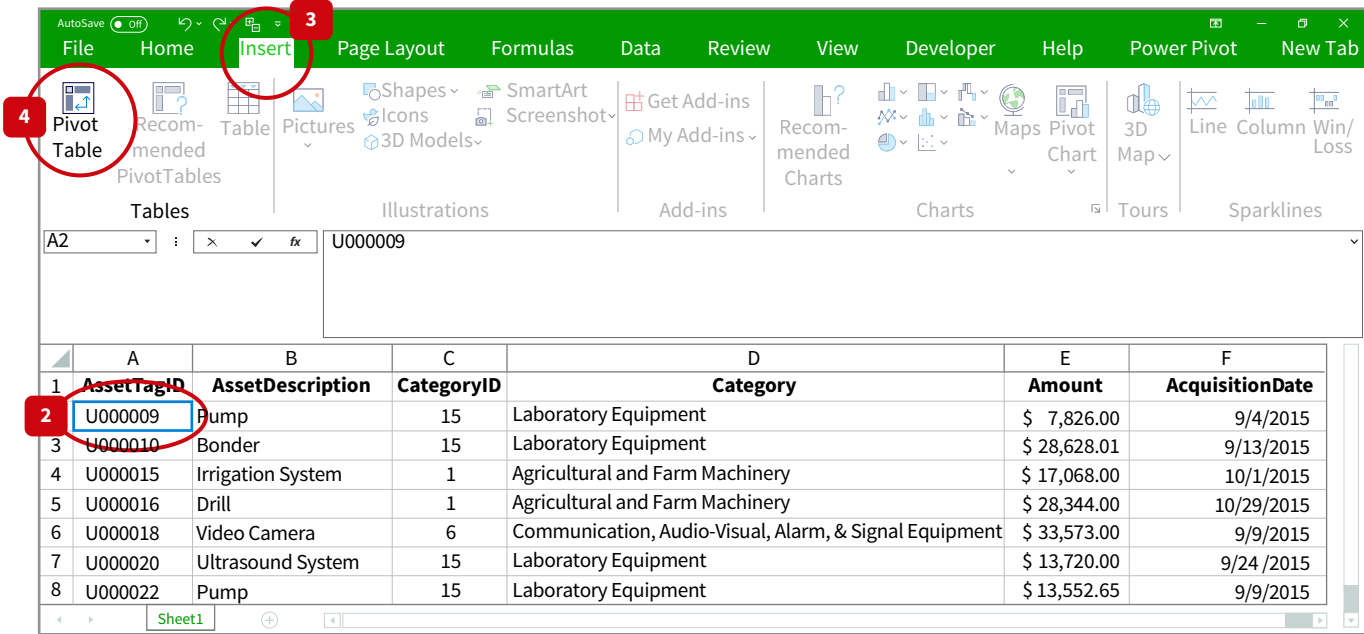
Once you know the basics of creating a pivot table in Microsoft Excel, you can use this tool to answer accounting questions.

Create a Microsoft Excel PivotTable

Follow these steps:

1. Open the spreadsheet with the data to summarize.
2. Click any cell within the data (the first cell in row A in [Illustration 2.17](#)).
3. Click the **Insert** option on the top menu ribbon.
4. A **PivotTable** input box option will appear at the top left of the screen ([Illustration 2.17](#)).

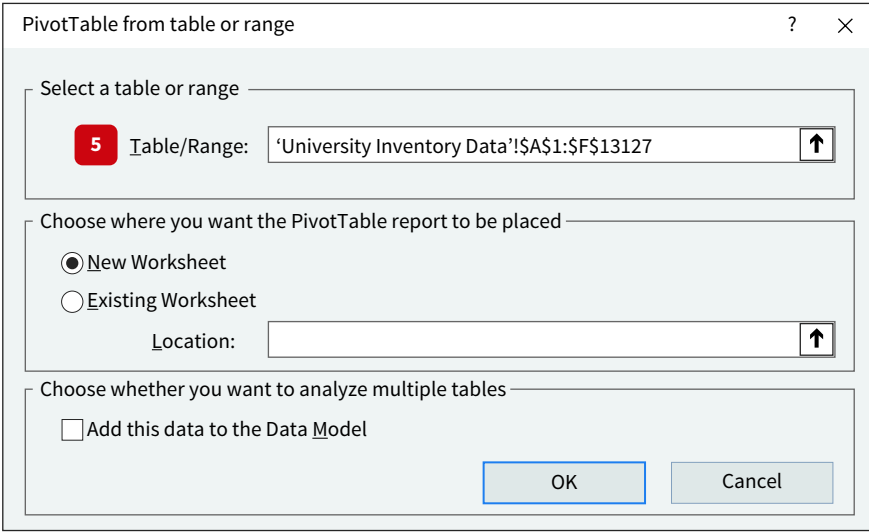
ILLUSTRATION 2.17 University Assets Data Excel Spreadsheet



When this option is selected a new dialogue box opens called **PivotTable from table or range** (Illustration 2.18).

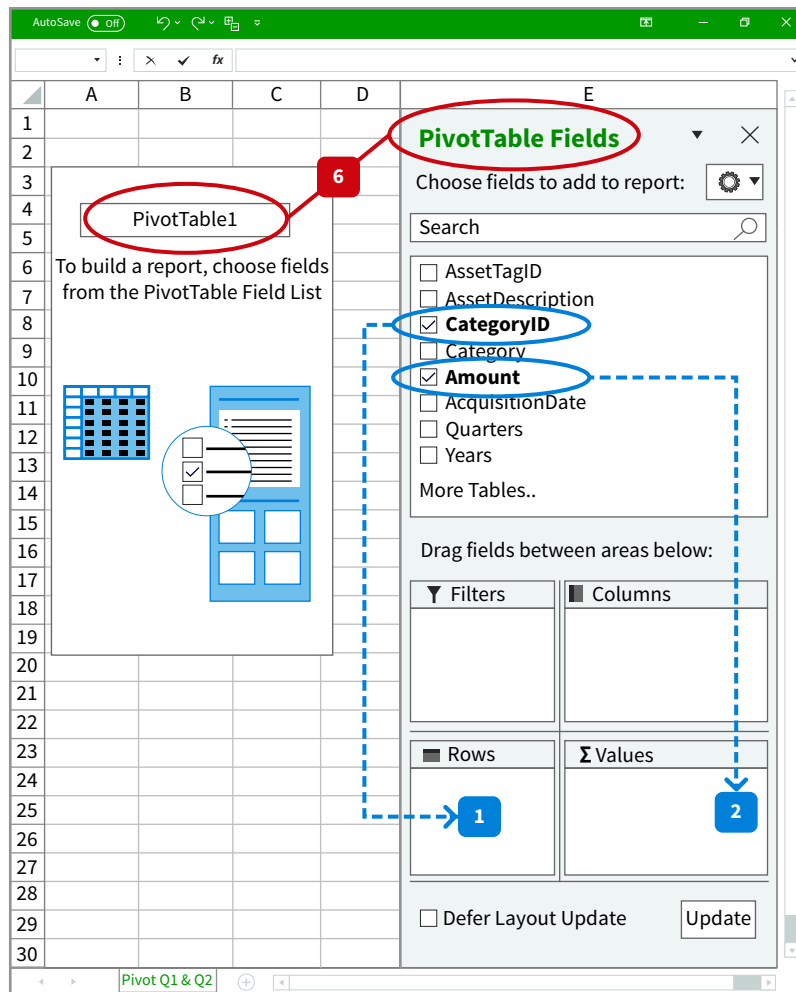
- 5. Ensure the Table/Range in the **Select a table or range** box reflects all data that should be included. (Remember to include the column headings.) Choose **New Worksheet** and click **OK** (Illustration 2.18).

ILLUSTRATION 2.18 Excel Create PivotTable Dialog Box



- 6. This will open a new spreadsheet. A blank PivotTable canvas will appear on the left, and the PivotTable Fields box that creates the PivotTable will be on the right (Illustration 2.19).

ILLUSTRATION 2.19 Blank PivotTable Canvas



Use the **PivotTable Fields** box to select what will appear in the PivotTable. Drag the column name to either rows, columns, values, or filters.

Data Let's use PivotTables to answer data questions about the university asset data. If you were verifying total costs for each of the asset categories, you may want to answer the following questions:

- Question 1: What is the total balance for assets in each category?
- Question 2: What is the total count of assets in each category?

Find the Total Balance by Category

Answer the first question by finding the total balance by category.

1. Drag **Category** to the **Rows** area.
2. Drag **Amount** to the **Values** area. The Σ Values in Columns is then automatically generated by Excel.
3. Excel will fill in the values for each category in the second column of the spreadsheet.

Illustration 2.20 shows the resulting PivotTable.

ILLUSTRATION 2.20 Excel PivotTable for Question 1: What is the Total Balance by Category?

Question 1	
Row Labels	3 Sum of Amount
Agricultural and Farm Machinery	\$ 3,838,203.58
Art Equipment	\$ 129,924.35
Athletic and Recreational Equipment	\$ 1,047,971.77
Boats, Boat Motors, and Marine Equipment	\$ 530,673.88
Cleaning and Maintenance Equipment	\$ 2,458,319.00
Communication, Audio-Visual, Alarm, and Signal Equipment	\$ 4,242,923.45
Computer Equipment	\$ 43,294,238.92
Drafting and Surveying Equipment	\$ 18,033.00
Electrical Appliances and Equipment	\$ 51,825.00
Electronic Equipment	\$ 278,233.02
Food Preparation and Serving Equipment	\$ 1,210,703.93
Furniture and Furnishings	\$ 3,060,781.37
Hand and Power Tools (Portable)	\$ 90,332.00
Industrial, Shop, and Construction Equipment	\$ 2,287,996.49
Laboratory Equipment	\$ 147,343,867.13
Material Handling Equipment	\$ 650,212.23
Medical Equipment	\$ 2,449,781.57
Miscellaneous Equipment	\$ 47,239.00
Motor Vehicles and Transportation Equipment	\$ 1,270,121.50
Musical Instruments	\$ 88,248.00
Office and Business Machines and Non-Consumable Supplies	\$ 1,587,495.45
Parking Control Equipment	\$ 230,506.72
Photographic Equipment	\$ 1,865,573.30
Plumbing, Heating, Air Conditioning, and Ventilating Equipment	\$ 146,953.15
Police, Fire Fighting, and Safety Equipment	\$ 102,458.55
Printing and Bookbinding Equipment	\$ 279,665.97
Refrigeration Equipment	\$ 6,467,000.41
Grand Total	\$ 225,069,282.74

4. Excel will default to a sum of the value. The type of measurement can be changed by clicking the down arrow in the **Sum of Amount** field and choosing the desired measure (Illustration 2.21).

ILLUSTRATION 2.21 Excel PivotTable Value Field Settings Dialog Box

Drag fields between

Filters

Rows

Category

Move Up

Move Down

Move to Beginning

Move to End

Move to Report Filter

Move to Row Labels

Move to Column Labels

Move to Values

Remove Field

Value Field Settings...

Count of Asset_T... ▼

☐ Defer Layout Update

Update

Value Field Settings

Source Name: AssetTagID

Custom Name: Count of AssetTagID

Summarize Values By

Show Values As

Summarize value field by

Choose the type of calculation that you want to use to summarize data from the selected field

Sum

Count

Average

Max

Min

Product

Number Format

OK

Cancel

Another useful feature of the Value Field Settings dialog box is the ability to change the number format in the PivotTable. (**Data** How To 2.1 at the end of this chapter explains the Show Values capability with pre-built calculations.)



Determine Total Count of Assets

To determine the total count of assets in each category (Question 2), use the first Excel PivotTable:

1. Drag **AssetTagID** to the **Values** area.
2. Click the down arrow, choose **Value Field Settings**, and choose **Count**.
The data can be summarized with several different calculations. The question asks about the number of assets per category, so choose **Count**.
3. Excel populates the next table column with the count by asset category.

The result is shown in **Illustration 2.22**.

ILLUSTRATION 2.22 Excel PivotTable for Question 2: What is the Total Count of Assets in Each Category?

Row Labels	Question 1 Sum of Amount	Question 2 Count of AssetTagID
Agricultural and Farm Machinery	\$ 3,838,203.58	171
Art Equipment	\$ 129,924.35	7
Athletic and Recreational Equipment	\$ 1,047,971.77	80
Boats, Boat Motors, and Marine Equipment	\$ 530,673.88	37
Cleaning and Maintenance Equipment	\$ 2,458,319.00	182
Communication, Audio-Visual, Alarm, and Signal Equipment	\$ 4,242,923.45	271
Computer Equipment	\$ 43,294,238.92	2543
Drafting and Surveying Equipment	\$ 18,033.00	2
Electrical Appliances and Equipment	\$ 51,825.00	5
Electronic Equipment	\$ 278,233.02	29
Food Preparation and Serving Equipment	\$ 1,210,703.93	112
Furniture and Furnishings	\$ 3,060,781.37	185
Hand and Power Tools (Portable)	\$ 90,332.00	5
Industrial, Shop, and Construction Equipment	\$ 2,287,996.49	154
Laboratory Equipment	\$ 147,343,867.13	8057
Material Handling Equipment	\$ 650,212.23	47
Medical Equipment	\$ 2,449,781.57	113
Miscellaneous Equipment	\$ 47,239.00	4
Motor Vehicles and Transportation Equipment	\$ 1,270,121.50	75
Musical Instruments	\$ 88,248.00	7
Office and Business Machines and Non-Consumable Supplies	\$ 1,587,495.45	83
Parking Control Equipment	\$ 230,506.72	12
Photographic Equipment	\$ 1,865,573.30	143
Plumbing, Heating, Air Conditioning, and Ventilating Equipment	\$ 146,953.15	17
Police, Fire Fighting, and Safety Equipment	\$ 102,458.55	12
Printing and Bookbinding Equipment	\$ 279,665.97	19
Refrigeration Equipment	\$ 6,467,000.41	754
Grand Total	\$ 225,069,282.74	13126

PivotTable Fields

Choose fields to add to report

Search

☒ AssetTagID
☐ AssetDescription
☐ CategoryID
☒ Category
☒ Amount
☐ AcquisitionDate
☐ Quarters
☐ Years
More Tables..

Drag fields between areas below:

Filters

Columns
Σ Values

Rows
Category

Σ Values
Sum of Amount
Count of AssetTagID

☐ Defer Layout Update
Update

Filtering Pivot Tables

One way to focus on a specific aspect of the data in a PivotTable is to use a filter. Applying a filter means that only data that fits its criteria will be displayed. There are three ways to filter in Excel:

- Apply the filter criteria to the Filter field area.
- Use AutoFilter in the PivotTable’s Row field.
- Insert one or more slicers.

Data Answering a third assets data question will demonstrate each filtering option:

Question 3: What is the total amount of purchases by category made in 2022?

Apply Filter Criteria to the Filter Field Box

To discover the total amount of purchases by category for the year 2022, create a PivotTable that shows assets purchased in 2022 by category:

1. Drag **Category** to **Rows**, and drag **Amount** to **Values**.
2. The focus is on assets purchased in 2022, so bring **Years** to the **Filters** area.
3. The word **Years** in the first column and the word **ALL** in the second column will now appear at the top of the PivotTable.
4. Clicking the down arrow next to **ALL** results in a dropdown box to select years to filter ([Illustration 2.23](#)).

ILLUSTRATION 2.23 How to Filter in an Excel PivotTable

The screenshot displays an Excel PivotTable and its associated task pane. The PivotTable has 'Years' as the filter field and 'ALL' as the selected filter. The task pane shows the 'Filters' area with 'Years' selected, the 'Rows' area with 'Category' selected, and the 'Values' area with 'Sum of Amount' selected. A red box highlights the 'Years' filter field, and a red arrow points to the 'ALL' dropdown arrow. A red box highlights the 'ALL' dropdown arrow, and a red arrow points to the '2022' option in the dropdown list. A red box highlights the '2022' option in the dropdown list, and a red arrow points to the '2022' option in the dropdown list. A red box highlights the '2022' option in the dropdown list, and a red arrow points to the '2022' option in the dropdown list.

Years	ALL
Agricultural and Farm Machinery	
Athletic and Recreational Equipment	
Boats, Boat Motors, and Marine Equipment	
Cleaning and Maintenance Equipment	
Communication, Audio-Visual, Alarm, and Signal E	
Computer Equipment	
Food Preparation and Serving Equipment	
Furniture and Furnishings	
Hand and Power Tools (Portable)	
Industrial, Shop, and Construction Equipment	
Laboratory Equipment	
Material Handling Equipment	
Medical Equipment	\$ 258,409.17
Miscellaneous Equipment	\$ 8,045.00
Motor Vehicles and Transportation Equipment	\$ 188,366.29
Office and Business Machines and Non-Consumable Supplies	\$ 103,113.50
Parking Control Equipment	\$ 23,274.20
Printing and Bookbinding Equipment	\$ 89,889.00
Refrigeration Equipment	\$ 389,043.31
Grand Total	\$ 20,869,919.66

Illustration 2.24 shows the result of filtering for assets purchased in 2022.

Years	2022
Row Labels	Sum of Amount
Agricultural and Farm Machinery	\$ 514,343.81
Athletic and Recreational Equipment	\$ 144,999.00
Boats, Boat Motors, and Marine Equipment	\$ 108,637.41
Cleaning and Maintenance Equipment	\$ 234,398.94
Communication, Audio-Visual, Alarm, and Signal Equipment	\$ 523,934.69
Computer Equipment	\$ 8,141,730.27
Food Preparation and Serving Equipment	\$ 77,915.86
Furniture and Furnishings	\$ 64,795.95
Hand and Power Tools (Portable)	\$ 49,665.00
Industrial, Shop, and Construction Equipment	\$ 117,352.04
Laboratory Equipment	\$ 9,825,228.22
Material Handling Equipment	\$ 6,778.00
Medical Equipment	\$ 258,409.17
Miscellaneous Equipment	\$ 8,045.00
Motor Vehicles and Transportation Equipment	\$ 188,366.29
Office and Business Machines and Non-Consumable Supplies	\$ 103,113.50
Parking Control Equipment	\$ 23,274.20
Printing and Bookbinding Equipment	\$ 89,889.00
Refrigeration Equipment	\$ 389,043.31
Grand Total	\$ 20,869,919.66

ILLUSTRATION 2.24 PivotTable Answer to Question 3: What is the Total Cost of Assets Purchased in 2022?

Use a Row Auto Filter

Filters can also be created in an Excel PivotTable with the Auto Filter functionality for rows. In this example, the question is how much was spent on computer equipment in 2022.

1. Instead of adding a category to the Filter area, click the down arrow in **Row Labels** to reveal other filtering choices.
2. **Illustration 2.25** shows the dropdown box that then appears. Here, select the categories of assets to filter.

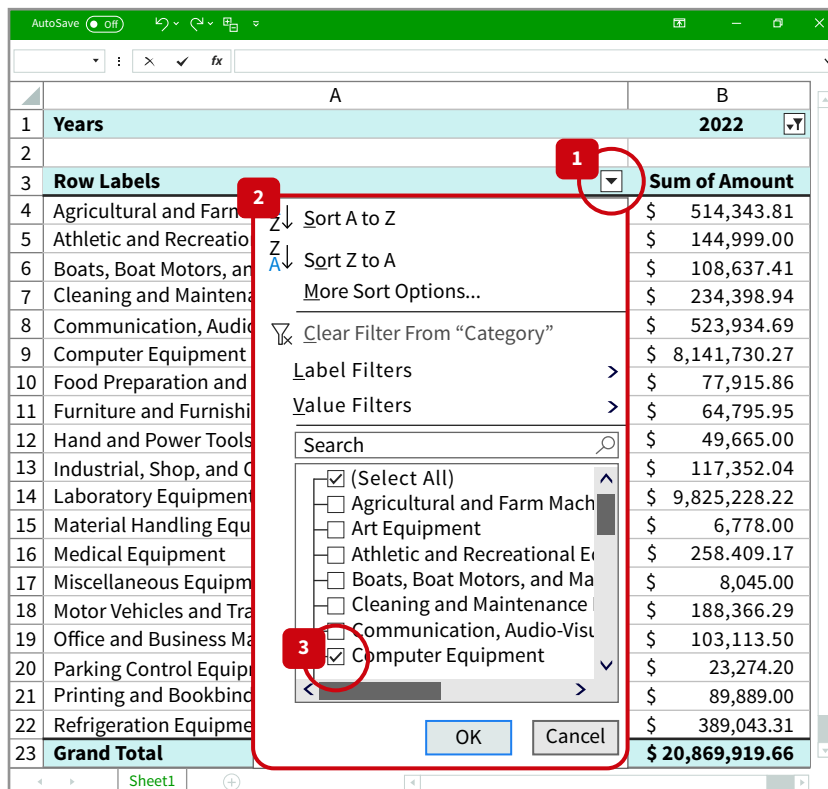
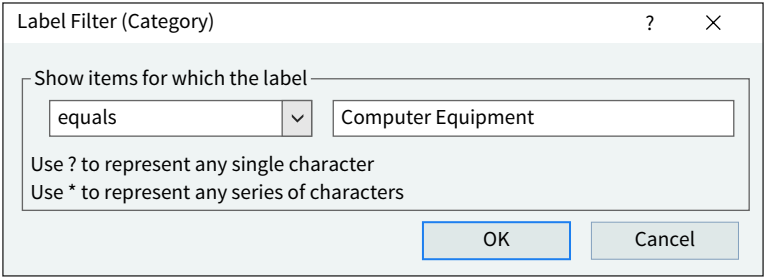


ILLUSTRATION 2.25 Excel PivotTable to Filter for Computer Equipment Only

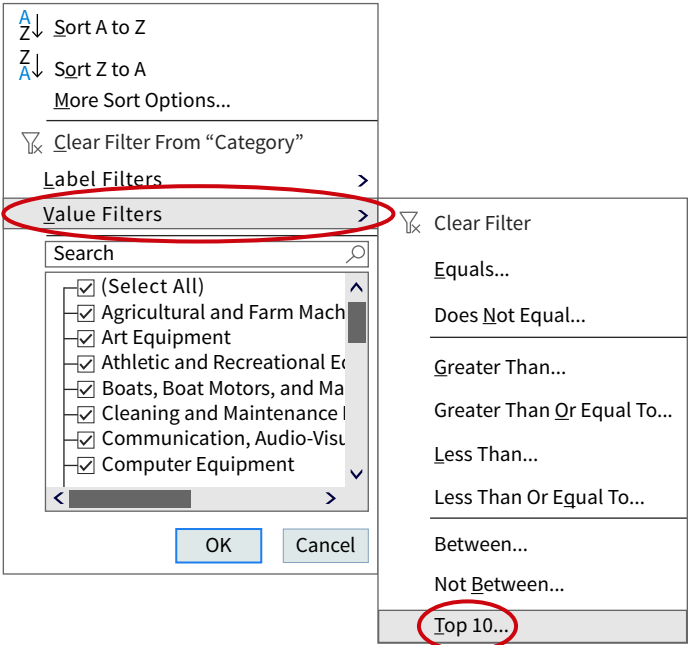
ILLUSTRATION 2.28 Excel PivotTable Label Filter Equals Dialog Box



What if the goal was to filter the data so only the top five asset categories appear?

- 1. Use the **Value Filters** and pick the option for **Top 10...** (Illustration 2.29).

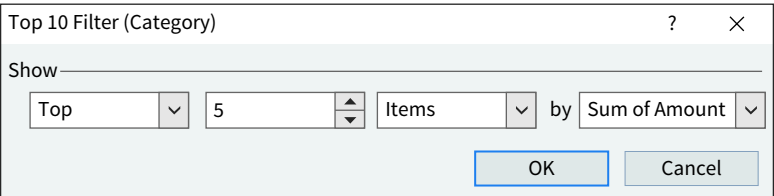
ILLUSTRATION 2.29 PivotTable Value Filter Options



- 2. A dialog box will open that allows changing the filter to top or bottom and the number of items to display.

In Illustration 2.30 the “Top 5” by “Sum of amount” was chosen.

ILLUSTRATION 2.30 PivotTable Value Filter For Top Values Input Box



The resulting Excel PivotTable is shown in [Illustration 2.31](#).

ILLUSTRATION 2.31 PivotTable Value Filter Results for Top 5 Amounts

Years	2022
Row Labels	Sum of Amount
Agricultural and Farm Machinery	\$ 514,343.81
Communication, Audio-Visual, Alarm, & Signal Equipment	\$ 523,934.69
Computer Equipment	\$ 8,141,730.27
Laboratory Equipment	\$ 9,825,228.22
Refrigeration Equipment	\$ 389,043.31
Grand Total	\$ 19,394,280.30

What about filtering for multiple dimensions at the same time?

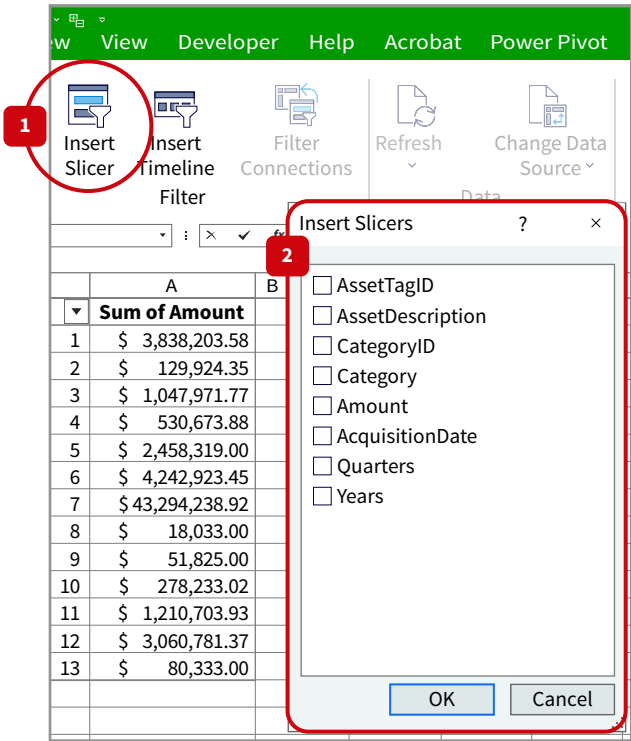
Use Slicers to Filter Data

Simultaneously filtering for multiple dimensions is often referred to as [slicing](#), or slice and dice, which is the process of breaking the data down into smaller parts or examining it from different viewpoints. [Slicers](#) are an analysis tool that separate the resulting analysis measures per the selected dimensions. All data analysis software has slicing capabilities:

- Microsoft Excel and Power BI use slicers.
- In Tableau, it is accomplished with interactive filters.

In all types of software, slicers customize interaction with a data set by providing a visual display of available filters. [Illustration 2.32](#) shows where the option to add slicers can be found in Microsoft Excel.

ILLUSTRATION 2.32 Adding Slicers to PivotTables



1. Click in the PivotTable and choose **Insert Slicer** from the menu. This will open a box with all the PivotTable fields.
2. Select the fields for the slicers. If the goal is to determine how much was spent on each category of assets by year, then slice data by asset category and year. To do this, select **Category** and **AcquisitionDate**.

Illustration 2.33 shows the resulting slicers. One is for category and the other is for year.

ILLUSTRATION 2.33 PivotTable Slicers Discovering the Total Amount by Category and Year

	A	B	C	D
1	Row Labels	Sum of Amount	Category	Years
2	Computer Equipment	\$ 8,141,730.27	Agricultural and Farm	2016
3	Grand Total	\$ 8,141,730.27	Athletic and Recreation	2017
4			Boats, Boat Motors, and	2018
5			Cleaning and Maintena	2019
6			Communication, Audio	2020
7			Computer Equipment	2021
8			Food Preparation and	2022
9			Furniture and Furnishin	2023
10				
11				
12				
13				
14				

In Illustration 2.33, clicking **Computer Equipment** in the **Category** slicer and **2022** in the **Years** slicer results in a PivotTable with the sum of computer equipment purchased in 2022.

Slicers can be added to Power BI by choosing the slicer tool in **Visualizations** (**Illustration 2.34**).

Filters

Search

Filters on this visual

Amount is (All)

Category is (All)

Add data fields here

Filters on this page

Add data fields here

Filters on all pages

Add data fields here

Visualizations

Slicer

Values

Category

Amount

Drill through

Cross-report

Off

Keep all filters

On

Add drill-through fields here

Fields

Search

Sheet1

AcquisitionDate

Amount

AssetDescription

AssetTagID

Category

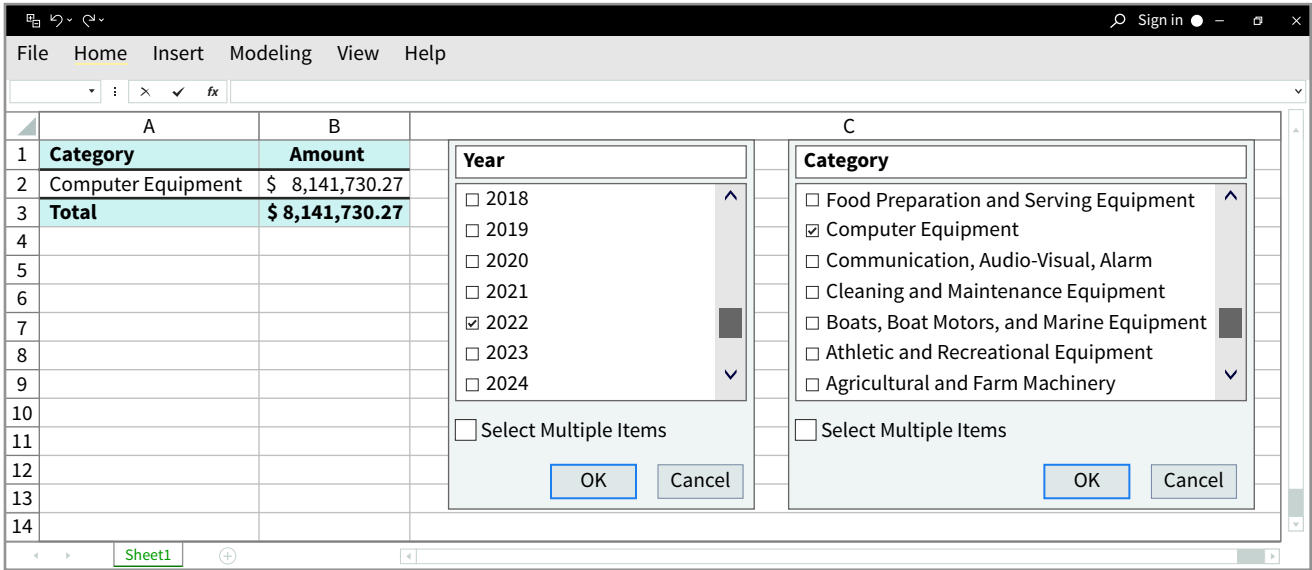
CategoryID

Year

ILLUSTRATION 2.34 Pivot Table Slicers Using PowerBI

Next, select **Category** to create category slicers and repeat for the AcquisitionDate slicers. **Illustration 2.35** shows the result.

ILLUSTRATION 2.35 Power BI Slicers for Category and Year



Because pivot table functions summarize data to quickly find answers to our questions, creating pivot tables is a core data analysis skill you will use frequently in your career.

Apply It 2.3

Analyze Sales with Excel PivotTables



Data **Financial Accounting** **Managerial Accounting** Your supervisor at Super Scooters has asked you to analyze sales for 2025 and revenue for all years. Create an Excel PivotTable to determine the following:

1. What are the total gross sales for each location in 2025?
2. What is the average gross sales amount for each model in 2025?
3. What is total revenue per year?

SOLUTION

1. Choose (or drag) the fields: **Location** for **Rows**, **Gross Sales** for **Values**, and **Year** for **Filters**. In the year filter pick 2025.

Year	2025
Row Labels	Sum of Gross Sales
Boston	\$ 1,433,578.00
Charlotte	\$ 1,653,506.00
Chicago	\$ 1,624,719.00
Dallas	\$ 1,134,785.00
Miami	\$ 1,606,576.00
Phoenix	\$ 1,376,503.00
Salt Lake City	\$ 1,593,184.00
Seattle	\$ 1,742,311.00
Grand Total	\$ 12,165,162.00

2. Begin by dragging the fields: **Model** to **Rows**, **Gross Sales** to **Values**, and **Year** to **Filters**. Change the Sum of Gross Sales to Average by opening **Value Field Settings** and choosing **Average**. In the year filter, select 2025.

Year	2025
Row Labels	Average of Gross Sales
Captain	\$ 15,855.84
Celeritas	\$ 4,977.54
Kicks	\$ 2,760.45
Lazer	\$ 7,981.45
Grand Total	\$ 9,098.85

3. Choose the following fields: **Years** to **Rows**, **Revenue** to **Values**.

Year	Sum of Total Revenue
2023	\$ 1,827,384.78
2024	\$ 4,086,546.19
2025	\$ 4,745,563.32
Grand Total	\$ 10,659,494.29

2.4 Which Descriptive Measures Help Us Understand Data?

LEARNING OBJECTIVE 4

Identify descriptive measures used to perform data analysis.

You have learned how to identify and retrieve data and some basic approaches to analyzing it. Earlier in the course, you also learned that there are four types of data analytics:

- Descriptive
- Diagnostic
- Predictive
- Prescriptive

Recall that descriptive analytics help reveal what has happened or is currently happening in the data. Why are descriptive analytics considered central to data analytics? Without that basic understanding of the data, advancing to more sophisticated data analysis methods is impossible. Sometimes descriptive analyses are all that are necessary, but more often this method is the precursor for diagnostic, predictive, and prescriptive analyses. The core data analysis skill for descriptive analytics is understanding descriptive statistics and correlation analysis.

Descriptive statistics reveal average observations in the data, the data's shape, and the distribution of the data. In addition, correlation analysis can show relationships in the data. Together, these statistics provide data insights.

Measures of Location

Measures of location determine the average, or typical, observation in the data set.

Mean, Median, and Mode

A **measure of central tendency** is a single value that describes a set of data by identifying the central position within that data set. There are three measures of central tendency:

- **Mean:** The sum of all observations in a data set divided by the total number of observations.
- **Median:** The middle value when the data is arranged from smallest to largest.
- **Mode:** The observation that occurs most frequently.

Determining the mean and median is the first step for understanding data during descriptive analysis. The two measures are often similar, but because the mean can be influenced by outliers (extreme values in the data set), there may be a large difference between them. If there are outliers in the data, then the median is a better representation of the central value in the data set.

Mode is useful in data sets with a small number of unique values. For example, an accounts receivable aging report may have values of 30, 60, and 90 days. A mode of the aging report data would reveal which category has the most observations. If there are few repeating values, then mode is not a useful measure of central tendency.

Excel is used here to calculate mean and median and interpret the results, but many tools can calculate mean and median values. In fact, all data visualization software can calculate mean and median. Regardless of how it is calculated, two things are important:

- Understanding how to *calculate* the measures.
- Knowing how to *interpret* the results.

Calculate Measures of Location

We will use the university example again to illustrate how to calculate the mean, median, and mode. Instead of assets, however, we will perform a descriptive analysis of the university payroll.

Data Illustration 2.36 is an excerpt from the university payroll data set. The data set shows the employee titles column and the annual salaries for all university employees (10,789 employees).

University Payroll Data

ILLUSTRATION 2.36 University Payroll Data

	B	C
1	Title	AnnualSalary
2	Program Coordinator	\$ 57,633.00
3	Eminent Scholar	\$ 94,626.00
4	Eminent Scholar	\$ 141,939.00
5	Assistant Professor	\$ 87,454.00
6	Network Administrator	\$ 48,874.00
10789	Associate Professor	\$ 23,998.00
10790	Professor	\$ 59,196.00
10791		

The first two steps involve Excel functions:

1. Calculate the mean using the Excel function AVERAGE. The formula is `=AVERAGE(C2:C10790)`. The result is \$40,065.88.
2. Calculate the median using the Excel function MEDIAN. The formula is `=MEDIAN(C2:C10790)`. The result is \$28,276.00.

Comparing the two measures shows a large difference between the mean and the median annual salary for an employee (\$11,789.88). What could cause this? Remember, mean can be affected by outliers. The data can be further examined to determine if there are extremely high or low salary amounts.

3. Finally, use the Excel filter option in the data file to filter the salaries from highest to lowest.

Illustration 2.37 shows the top five salaries and the bottom five salaries.

Top Filter: Top Five University Salaries

ILLUSTRATION 2.37 University Salaries Highest to Lowest

	A	B	C
1	EmployeeNumber	Title	AnnualSalary
2	1853	Professor	\$ 468,675.00
3	2227	Professor	\$ 426,091.00
4	7814	Professor	\$ 425,000.00
5	9964	Professor	\$ 406,909.00
6	6166	Professor	\$ 397,118.00

Bottom Filter: Bottom Five University Salaries

	A	B	C
1	EmployeeNumber	Title	AnnualSalary
2	1674	Publication Graphic Artist	\$ 172.00
3	3031	Graduate Research Assistant	\$ 159.00
4	5008	Graduate Assistant	\$ 152.00
5	4766	Graduate Assistant	\$ 123.00
6	6845	Professor	\$ 78.00

There is a large difference between the highest paid employee and the lowest paid employee

Given how much higher the mean salary is compared to the median, it is likely there are more salaries at lower amounts than at higher amounts. We can better understand the difference between the mean and median salary amounts by considering how much variation is in the data.

Measures of Dispersion

In the salary example there was a large difference between the mean and median annual salary amounts. **Measures of dispersion**, which describe the amount of variation in the data, can help find the cause of this disparity. Are the data spread out or are they compact? In other words, how far apart are all the observations, or data points, from the mean?

Variance and Standard Deviation

There are two widely-used measures of dispersion:

- **Variance** is the average squared distance between the data points in the data set and the mean.
- **Standard deviation** is the square root of the variance.

Although variance is necessary to calculate standard deviation, only standard deviation is usually reported because it is easier to interpret than variance. It is easier to understand because it is in the same units as the mean. In the university payroll example, standard deviation would be in dollars of annual salary.

Calculate Measures of Dispersion

Data Measures of dispersion can be calculated using Microsoft Excel (**Illustration 2.38**).

ILLUSTRATION 2.38 University Payroll Data

University Payroll Data		
	B	C
1	Title	AnnualSalary
2	Program Coordinator	\$ 57,633.00
3	Eminent Scholar	\$ 94,626.00
4	Eminent Scholar	\$ 141,939.00
5	Assistant Professor	\$ 87,454.00
6	Network Administrator	\$ 48,874.00
10789	Associate Professor	\$ 23,998.00
10790	Professor	\$ 59,196.00
10791		
10792	Variance	=VAR(C2:C10790)
10793	Standard Deviation	=STDEV(C2:C10790)

Again, calculate these measures by performing calculations in Excel:

1. Calculate the variance using the Excel function VAR. The formula is =VAR(C2:C10790). The result is \$1,939,797,496.92
2. Next, calculate the standard deviation using the Excel function STDEV. The formula is =STDEV(C2:C10790). The result is \$44,043.13

While there is not a practical interpretation of the variance number, there is for standard deviation:

- A low standard deviation indicates the observations in the data set tend to be close to the mean of the data set.
- A high standard deviation indicates the values are spread out over a wider range.

In this example a standard deviation of \$44,043.14 indicates that an observation, in this case an employee's annual salary, in the data set could vary by \$44,043.14 from the mean. Recall that the mean is \$40,068.88, so \$44,043.14 would be considered a high standard deviation.

Another way to evaluate dispersion is to use a scatterplot (also called a scatter chart) to visualize the data ([Illustration 2.39](#)).

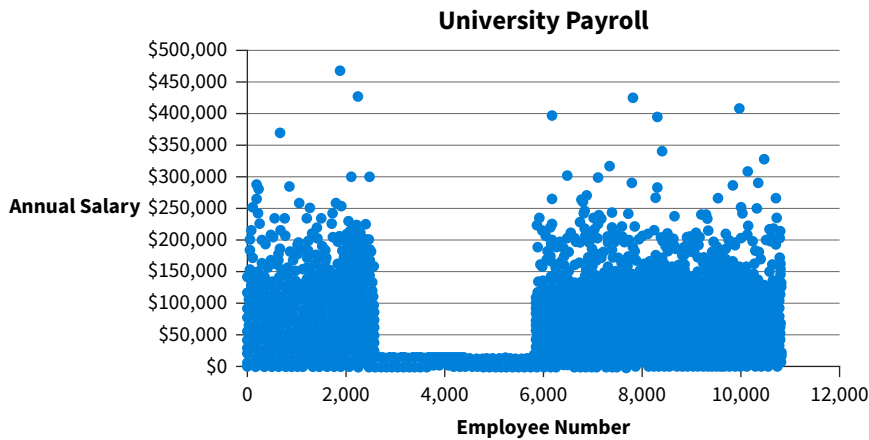


ILLUSTRATION 2.39 University Payroll Scatterplot

Salaries are widely dispersed between \$0 and a little over \$450,000. This visualization of the data coincides with the large standard deviation of \$44,043.14

Measures of Shape

Besides understanding the dispersion of data, it is also important to understand its distribution, or shape. **Measures of shape** describe the distribution of the data in the data set. How a data set is shaped can reveal the best measure of central tendency to use, or it can show patterns in the data.

Skewness and Kurtosis

Data sets are either symmetrical or asymmetrical in shape. In a symmetrical distribution, the mean, median, and mode are equal and the data distribution to the right of the mean mirrors the data to the left of the mean. A symmetrical distribution will look like a bell curve in a graph ([Illustration 2.40](#)).

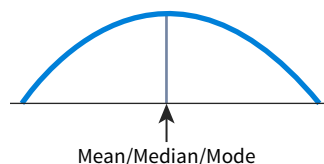


ILLUSTRATION 2.40 Symmetrical Distribution

The distribution's shape can be determined by graphing the data or by using two statistical measures. Often, both are appropriate. Two measures of shape are skewness and kurtosis.

Skewness describes the lack of symmetry of data:

- Distributions that tail off to the right of the mean are considered positively skewed.
- Distributions that tail to the left of the mean are negatively skewed.

ILLUSTRATION 2.41 Skewness Examples

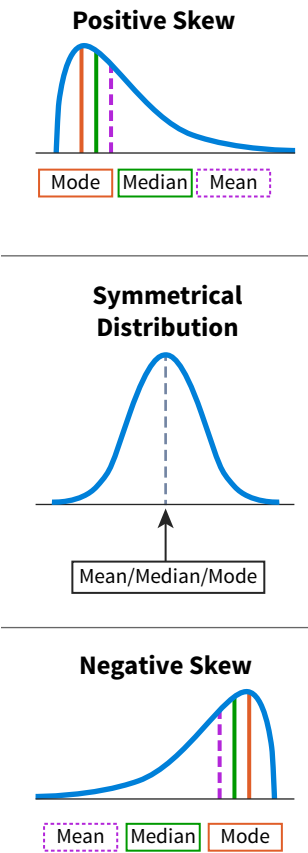


Illustration 2.41 shows a positive, symmetrical, and negative skewed distribution. The **coefficient of skewness (CS)** measures the skewness of a distribution. If the CS is negative, then the data are left skewed (tails off to the left). If the CS is positive, the data are right skewed (tails off to the right). The degree of skewness can be interpreted by the absolute value of the CS:

- $|CS| > 1$, high degree of skewness
- $0.5 \leq |CS| \leq 1$, moderate skewness
- $|CS| < 0.5$, relative symmetry

Calculate the coefficient of skewness using the Excel function `=SKEW(data range)`. **Data** Illustration 2.42 shows the Excel function and result using the university payroll data file. The CS is a positive 2.3, indicating the data have a high degree of skewness and skew to the right.

ILLUSTRATION 2.42 Skewness and Kurtosis of University Payroll Data

Measure	Function	Result
CS	<code>=SKEW(C2:C10790)</code>	2.3
CK	<code>=KURT(C2:C10790)</code>	8.2

Kurtosis refers to how peaked or flat a histogram of the data is. The **coefficient of kurtosis (CK)** measures the degree of kurtosis of a distribution:

- If the CK is greater than three, then the data are somewhat peaked with less dispersion.
- If the CK is less than three, the data are somewhat flat with a wide degree of dispersion.

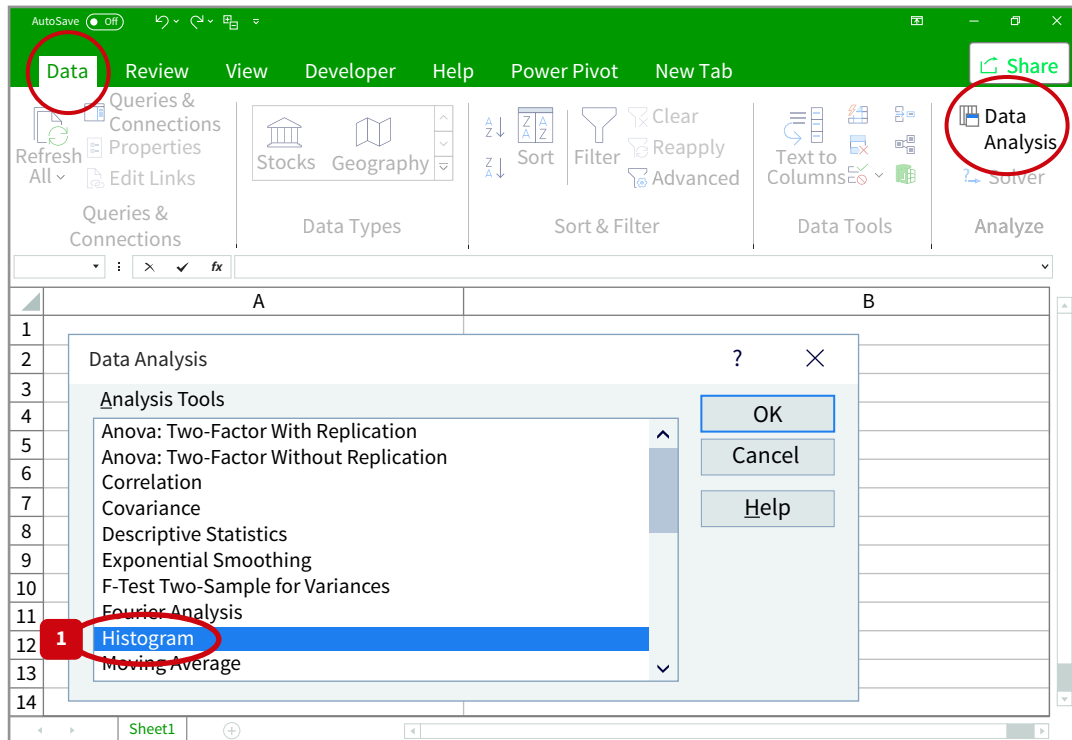
The coefficient of kurtosis can be calculated in Excel using the function `=KURT(data range)`. Illustration 2.42 shows the CK of the university salary data is 8.2. This indicates the data are peaked with less dispersion.

Frequency Distributions and Histograms

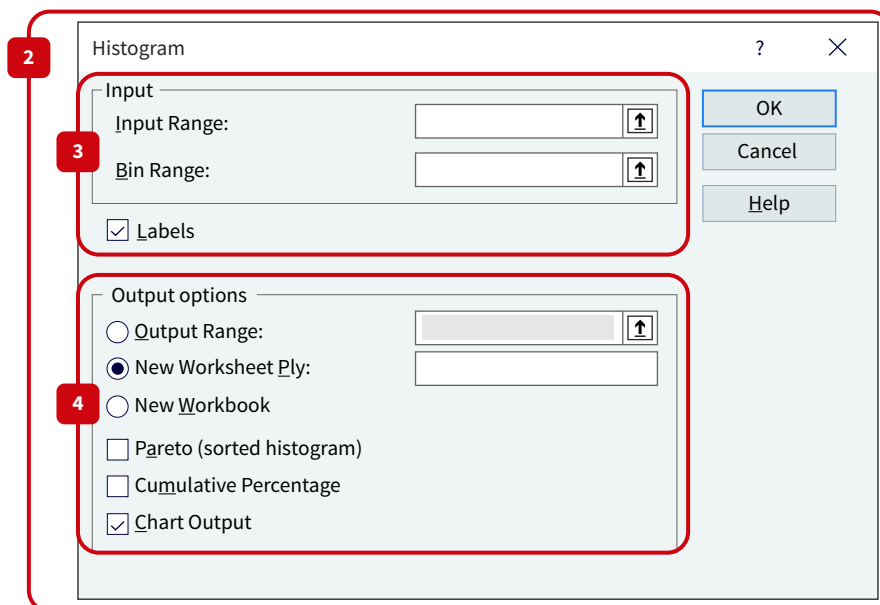
Along with the CS and CK measures, the shape of a distribution can be visualized with two other measures:

- A **frequency distribution** is a representation of the data that summarizes the number of observations within a given interval. For example, the number of employees by salary dollar groupings.
- A **histogram** is a bar chart of a frequency distribution in which the height of the bar reflects the frequency within the interval. For example, we may want to group salaries by dollar amounts of \$10,000, \$20,000, and \$30,000 and then count how many employees are in each category.

These measures can be created in data visualization software such as Power BI, Tableau, and in Microsoft Excel using the Analysis Toolpak. Analysis Toolpak is a free add-on for Excel. Once added, it will be under the **Data** tab. Illustration 2.43 shows where to find the Data Analysis tool and the corresponding box that opens after clicking **Data Analysis**.

ILLUSTRATION 2.43 Excel Data Analysis Tool

1. To prepare a histogram and frequency distribution, choose **Histogram** and click **OK**.
2. A dialog box opens ([Illustration 2.44](#)).
3. There are two input boxes:
 - The **Input Range** is the data that will be visualized. It is helpful to select the column title and check the **Labels** box so the histogram is labeled with the title of the data being visualized ([Illustration 2.44](#)).
 - The next input box is **Bin Range**, which defines the groups used for the frequency distribution and histogram bars. These must be established before opening the dialog box. Create the Bin Range by setting up a column of the values in which the data should be grouped. In general, the values should be in equal increments and the number of groups should be between five and fifteen.
4. Finally, choose where the outputs should go. Be sure to check the **Chart Output** box before clicking **OK**.

**ILLUSTRATION 2.44** Histogram Dialog Box

How do you know how many groups to create? The best way is to experiment with the data to find the number that will create a useful visualization. For example, using fewer groups means the group widths will be wider and the visualization may provide less detailed insight. It is often helpful to divide the range (maximum observation – minimum observation) by the number of groups you would like display as a starting point.

Data These steps can be used to create a histogram with the university payroll data file:

- 1. Open the spreadsheet and create a column for the bins.
- 2. First, determine the range of the data. The minimum salary is \$78.00 and the maximum salary is \$468,675.00, indicating there is a large range of data. Create 12 bins, starting with \$8,000 in \$20,000 increments (**Illustration 2.45**).

ILLUSTRATION 2.45 Payroll Data, Bins, and Histogram Dialog Box

Payroll Data, Bins, and Histogram Dialog Box

	B	C	D	E
1	Title	Annual Salary		Bins
2	Program Coordinator	\$ 57,633.00		8,000
3	Eminent Scholar	\$ 94,626.00		28,000
4	Eminent Scholar	\$ 141,939.00		48,000
5	Assistant Professor	\$ 87,454.00		68,000
6	Network Administrator	\$ 48,874.00		88,000
7	Student Program Coordinator	\$ 42,630.00		108,000
8	Professor	\$ 113,736.00		128,000
9	Professor	\$ 2,170.00		148,000
10	Professor	\$ 115,228.00		168,000
11	Research Assistant	\$ 6,700.00		188,000
12	Research Assistant	\$ 6,700.00		208,000
13	Research Assistant	\$ 20,100.00		228,000
14	Professor	\$ 119,900.00		
10789	Associate Professor	\$ 23,998.00		
10790	Professor	\$ 59,196.00		

Histogram

Input

Input Range:

Bin Range:

☒ Labels

Output options

☐ Output Range:

☒ New Worksheet Ply:

☐ New Workbook

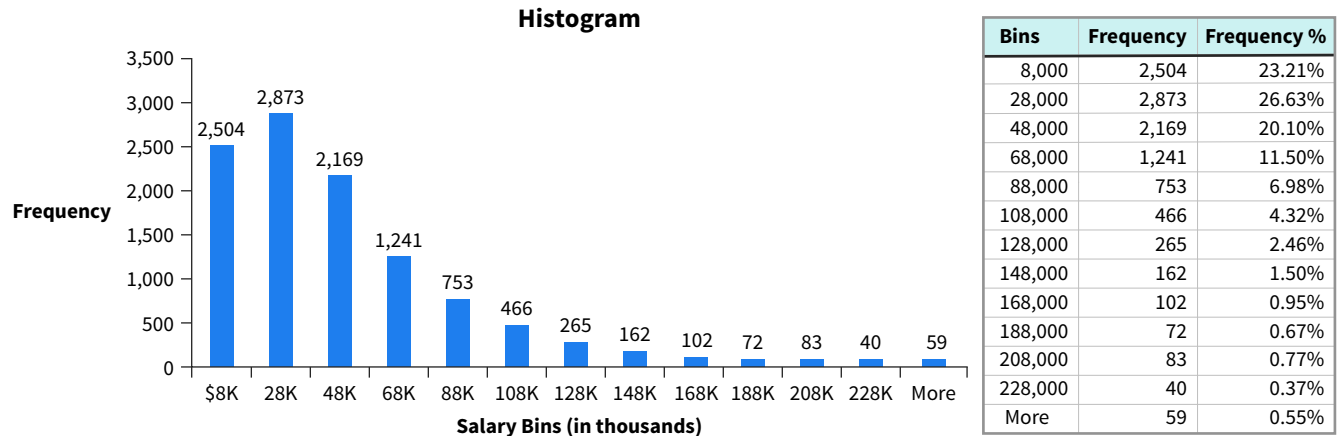
☐ Pareto (sorted histogram)

☐ Cumulative Percentage

☒ Chart Output

OK Cancel Help

The results from the histogram dialog box are presented in **Illustration 2.46**.

ILLUSTRATION 2.46 University Payroll Data Histogram and Frequency Distribution

Note that the *Bins* and *Frequency* are generated by Excel when the histogram is created. The column for *Frequency %* was calculated by dividing the *Frequency* for each bin by the total number of observations (10,789).

The histogram reveals some information about the shape of the data set, including that most of the data is grouped between \$8,000 and \$48,000:

- The data then trails off to the right. This supports the results of the measures of location and shape. Recall the median for the data is \$28,276.00, and the mean is \$40,065.88. The data peaks at the mode.
- The data then tails off to the right, which supports the result for the coefficient of skewness of a positive 2.3.
- Finally, the data are highly peaked. The majority (69%) are within the first three bars on the histogram. This supports the result for the coefficient of kurtosis of 8.2 that indicates highly peaked data with less dispersion.

Descriptive Statistics Tools

You have now learned how to calculate measures of location, dispersion, and shape using single Excel functions. There is another Excel tool, Descriptive Statistics, that calculates all these measures at once (**Illustration 2.47**):

1. Choose **Descriptive Statistics** from the **Analysis Tools** list.
2. Use the dialog box for **Descriptive Statistics** to input the data range to analyze.
3. Once the data range is entered, select labels (if you chose the row with the column title), choose where the output will go, select **Summary statistics**, and click **OK**.

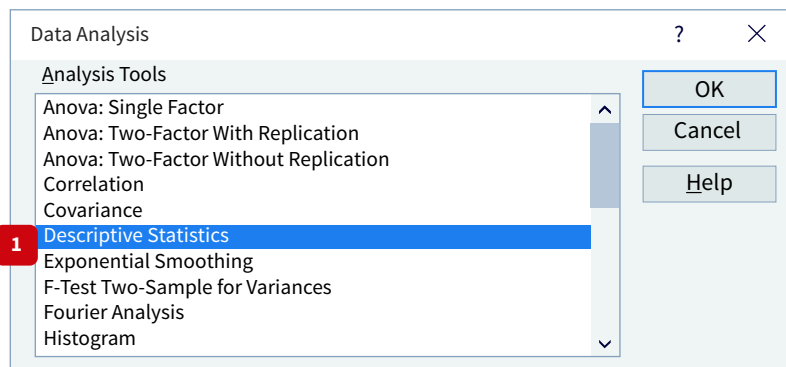
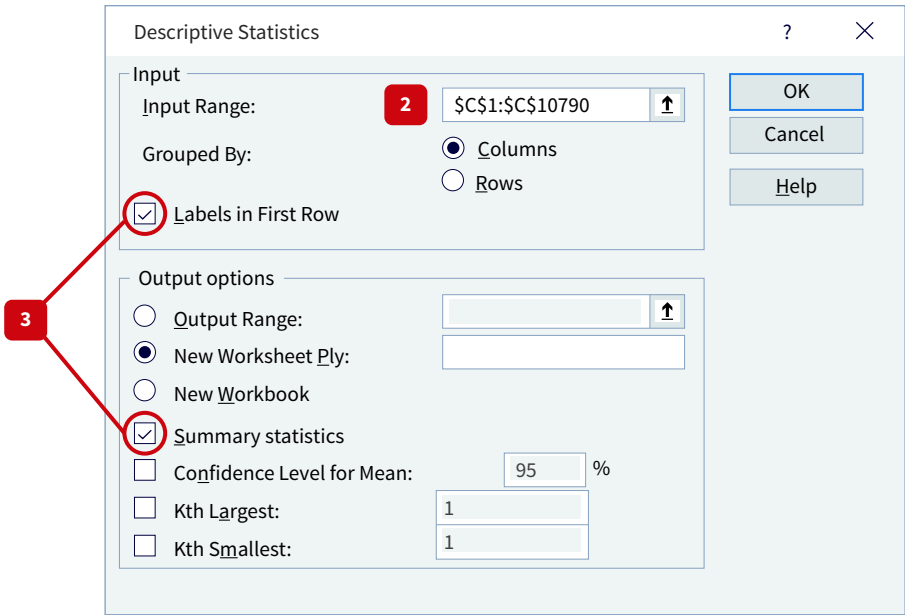
**ILLUSTRATION 2.47** Descriptive Statistics Excel Tool and Dialog Box

ILLUSTRATION 2.47 (Continued)



Excel will then calculate the descriptive statistics and print the results on a new worksheet (Illustration 2.48).

ILLUSTRATION 2.48 University Salary Data Descriptive Statistics

Annual Salary		
Mean	\$	40,065.88
Standard Error	\$	424.02
Median	\$	28,276.00
Mode	\$	3,269.00
Standard Deviation	\$	44,043.13
Sample Variance	\$	1,939,797,496.92
Kurtosis	\$	8.19
Skewness	\$	2.30
Range	\$	468,597.00
Minimum	\$	78.00
Maximum	\$	468,675.00
Sum	\$	432,270,823.00
Count		10,789.00

Statistical software packages typically have a similar feature to calculate multiple descriptive statistics simultaneously. Whether we use multiple single Excel functions or just one to calculate it, revealing the shape of a data set helps us better understand the data. There is also another important component—understanding relationships within a data set.

Correlation Analysis

Correlation analysis can reveal relationships in data by measuring the linear relationship between two variables. The first step is understanding how variables are correlated, and the second step involves calculating the correlation.

Interpret Correlation Coefficients

Linear correlation of continuous variables is measured by the **correlation coefficient**, also known as the Pearson Product Moment Correlation Coefficient. This measure is a numerical value between -1 and $+1$. The higher the absolute number, the greater the strength of the relationship.

A correlation can be negative, zero, or positive (**Illustration 2.49**):

- A negative correlation is an inverse relationship. As one variable increases, the other decreases. There is a negative relationship between soup sales and temperature because as temperatures decrease, the soup sales increase.
- No correlation indicates no relationship between the variables. We would not expect the outdoor temperature to have an impact on the sale of cereal, for example.
- A positive correlation coefficient indicates that as one variable increases, so does the other variable. We would expect a positive relationship between ice cream sales and outdoor temperature. As the temperature rises, ice cream sales tend to rise as well.

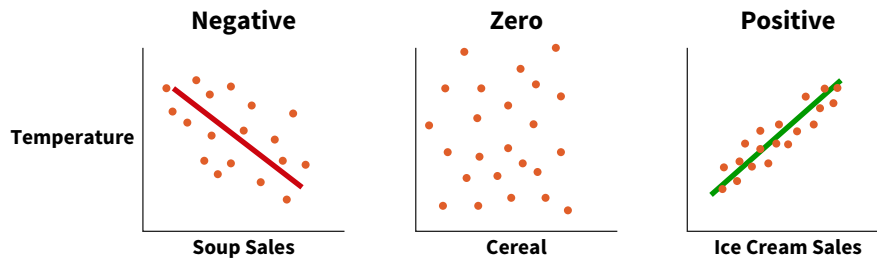


ILLUSTRATION 2.49 Correlation Examples

We can also consider the strength of a relationship. The higher the correlation coefficient, between a negative 1 and a positive 1, the stronger the correlation. **Illustration 2.50** is a guide to determining if a correlation coefficient indicates a weak, moderate, or strong relationship.

Correlation r	Interpretation
Exactly -1	A perfect negative linear relationship
-0.70	A strong negative linear relationship
-0.50	A moderate negative linear relationship
-0.30	A weak negative linear relationship
0	No linear relationship
$+0.30$	A weak positive linear relationship
$+0.50$	A moderate positive linear relationship
$+0.70$	A strong positive linear relationship
Exactly $+1$	A perfect positive linear relationship

ILLUSTRATION 2.50 Interpreting Correlation Coefficients

Imagine examining the relationship between ice cream sales and the outdoor temperature. If the correlation coefficient of sales and temperature was positive 0.75, then there is a strong positive relationship between sales and temperature. As temperatures rise, sales of ice cream increase and vice versa. (**Illustration 2.51**)

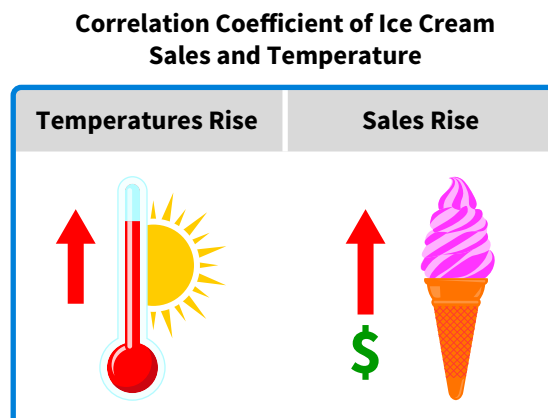
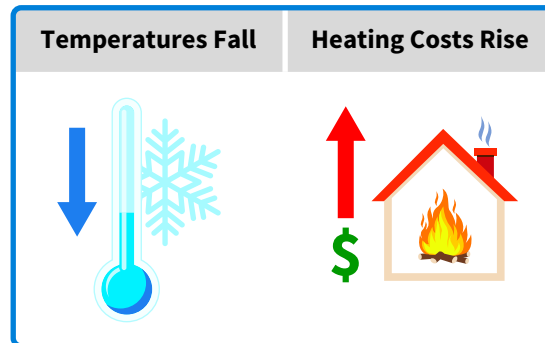


ILLUSTRATION 2.51 Positive Correlation Example

Suppose instead we are examining the relationship between heating costs and temperature, and the correlation coefficient is a negative 0.70 (**Illustration 2.52**)

ILLUSTRATION 2.52 Negative Correlation Example

Correlation Coefficient of Heating Costs and Temperature



In this case, there is a strong negative correlation between heating costs and temperature. As the temperature decreases, the cost of heating a home increases and vice versa.

Perform Correlation Analysis

Correlation can be visually assessed by preparing a scatterplot, like those in Illustration 2.49, and then plotting a line. This will indicate if a correlation exists and whether it is positive or negative.

The correlation coefficient can be calculated by hand with a formula or with software. All statistical software can calculate a correlation coefficient, including Microsoft Excel, which is used in this example. Regardless of the software used, the interpretation of the results is the same. There are two ways to perform correlation analysis in Excel:

- Using the CORREL function.
- Using the Correlation option in the Data Analysis tool.

The benefit of using the Correlation option in the Data Analysis tool is that a correlation table for more than one variable can be created at the same time. **Data** We use the university payroll data to illustrate the correlation option in Excel and then interpret the results.

In the Salary Hours tab in the data file (**Illustration 2.53**), perform a correlation analysis to see if there is a correlation between annual salary and hours worked.

ILLUSTRATION 2.53 University Salary and Hours Data

University Salary and Hours Data

AutoSave Off    

	F	G	H
	Title	AnnualSalary	HoursWorked
1			
2	Program Coordinator	\$ 57,633.00	2,040
3	Eminent Scholar	\$ 94,626.00	612
4	Eminent Scholar	\$ 141,939.00	918
5	Assistant Professor	\$ 87,454.00	1,530
6	Network Administrator	\$ 48,874.00	2,040
7	Student Program Coordinator	\$ 42,630.00	2,040
10789	Associate Professor	\$ 23,998.00	387.6
10790	Professor	\$ 59,196.00	306

Sheet1

Illustration 2.54 shows the correlation dialog box that opens after clicking **Correlation** and selecting **OK** in the **Data Analysis Tools** dropdown menu. The input range in the dialog

box contains the columns being tested for correlation. This example is testing for correlation between column G (Annual Salary) rows 1 – 10790, and column H (Hours Worked) rows 1 – 10790.

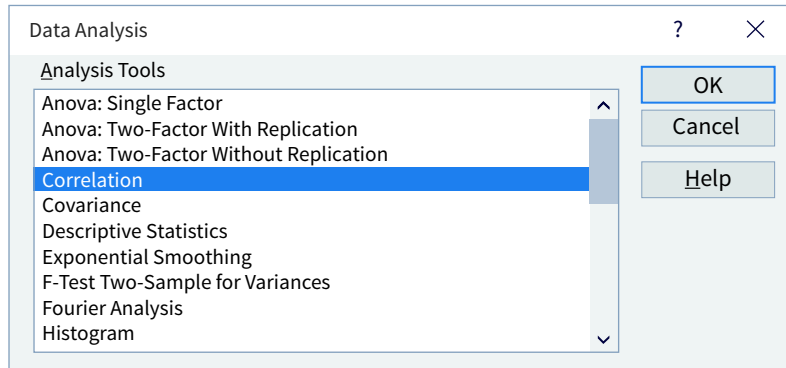
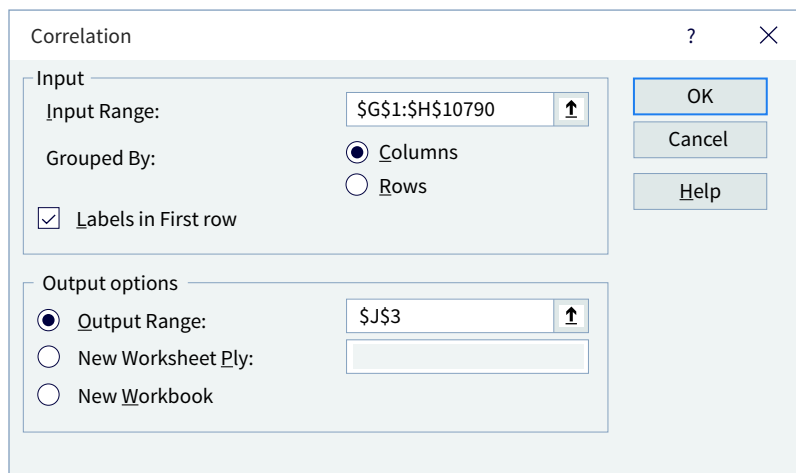


ILLUSTRATION 2.54 Data Analysis Tool and Correlation Dialog Box



The dialog box in Illustration 2.54 shows the inputs necessary to run the correlation analysis. The result of the correlation analysis is shown in **Illustration 2.55**:

- The correlation coefficient is 0.552.
- There is a moderate positive correlation between annual salary and hours worked.

In other words, as hours worked increases so does annual salary.

	Annual Salary	Hours Worked
Annual Salary	1	
Hours Worked	0.552	1

ILLUSTRATION 2.55 University Salary Correlation Analysis

Why is there not a stronger correlation? A strong correlation between the hours worked and the amount of annual salary paid seems to make sense. However, if employees are paid a fixed annual salary rather than hourly that would reduce the correlation to hours.

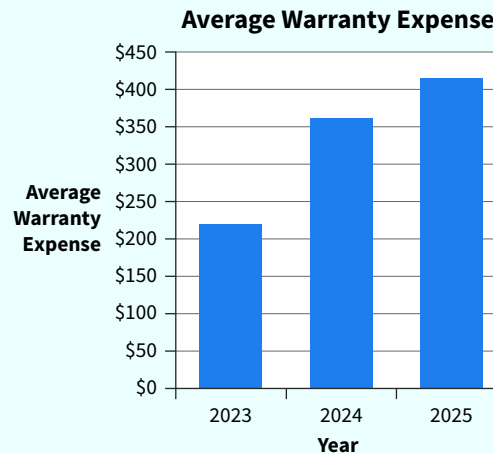
Correlation analysis can help uncover relationships in the data and understand their strength. But never assume that one variable causes a change in the other because there is correlation between the two. It is possible to find a correlation between two variables that have nothing to do with one another. This is a spurious correlation, which occurs when there is a mathematical, but not a logical, relationship between two variables. Always ensure correlations make sense before using one to make decisions.

Apply It 2.4

Use Descriptive Statistics to Audit Warranty Expense



Data Auditing As an external auditor, you have been assigned to the audit engagement for Super Scooters. One of your responsibilities is to review warranty expense. As the illustration shows, average warranty expense for Super Scooters has been rising over the last three years.



To analyze the warranty expense, you have decided to use descriptive statistics. Perform the following analyses and interpret your results.

1. Use the Descriptive Statistics option in the Data Analysis tool to calculate the descriptive statistics for warranty expense from 2023 to 2025. Interpret the following measures:
 - Mean
 - Median
 - Standard deviation
 - Kurtosis
 - Skewness
2. Perform descriptive statistics for warranty expense in 2025. Interpret the following measures:
 - Mean
 - Median
 - Kurtosis
 - Skewness
3. Prepare a histogram of 2025 Warranty Expense with the following bins: 200, 400, 600, 800, 1000, 1200, and 1,400.
 - Chart the data.
 - Does the shape and distribution of your histogram support the kurtosis and skewness measures?

SOLUTION

1. Descriptive statistics for warranty expense from 2023 to 2025:

Total Warranty	
Mean	343.57
Standard Error	4.06
Median	300
Mode	240
Standard Deviation	244.90
Sample Variance	59,977.92
Kurtosis	1.54
Skewness	1.23
Range	1,493
Minimum	7
Maximum	1,500
Sum	1,252,326
Count	3,645

Measure Interpretation:

Measure	Result	Interpretation
Mean	The mean warranty cost for the 3-year period was \$343.57.	The average warranty cost for all 3,645 sales over the 3-year period is \$343.57.
Median	The median warranty cost for the 3-year period was \$300.00.	The middle value of warranty expense for the 3-year period, when ranking warranty expense from high to low, is \$300.
Standard deviation	The standard deviation over the 3-year period was \$244.90.	This is a high standard deviation compared to the mean and median. It indicates a wide dispersion of the warranty costs. For any given sale, the warranty cost could be + or – \$244.90 from the mean.
Kurtosis	The coefficient of kurtosis is 1.54.	This is less than 3, indicating that the shape of the distribution is somewhat flat with a wide degree of dispersion.
Skewness	The coefficient of skewness is a positive 1.23.	This is greater than 1, indicating that the data peaks around the mean and then tails off to the right.

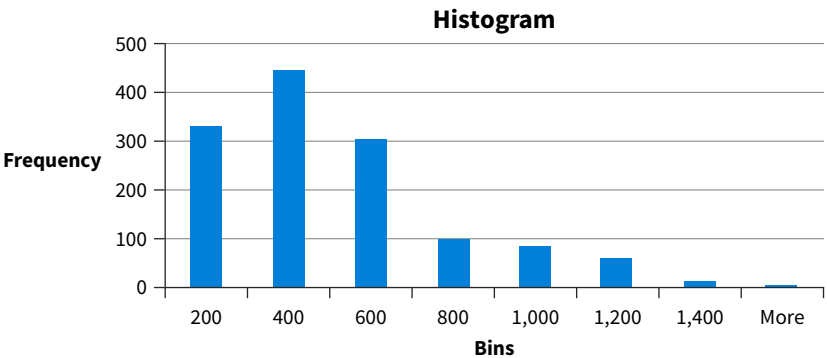
2. Descriptive statistics for warranty expense in 2025:

Total Warranty-2025	
Mean	414.18
Standard Error	7.551634
Median	330
Mode	330
Standard Deviation	276.13
Sample Variance	76,245.34
Kurtosis	0.93
Skewness	1.18
Range	1,460
Minimum	40
Maximum	1,500
Sum	553,759
Count	1,337

Measure Interpretation:

Measure	Result	Interpretation
Mean	The mean warranty cost for 2025 was \$414.18.	This is higher than the three year average found in question 1.
Median	The median warranty cost for 2025 was \$330.00.	This represents the exact middle of the distribution if the data is arranged from lowest to highest.
Kurtosis	The coefficient of kurtosis is 0.93.	This is less than three, indicating that the shape of the distribution is somewhat flat with a wide degree of dispersion.
Skewness	The coefficient of skewness is a positive 1.18.	This is greater than one, indicating the data peaks around the mean and then tails off to the right.

3. Histogram:



Yes, the kurtosis indicates a wide dispersion. The warranty expense is spread out from \$200 to \$1,200:

- There are three peaks versus one high peak. The skewness indicates the data peaks around the mean and then tails off to the right.
- The highest peak is around the mean of \$414 and then tails off to the right.

2.5 How is Visualization Used in Data Analysis?

LEARNING OBJECTIVE 5

Summarize how data visualization explores and explains data.

Data visualization is the graphical presentation of data and information. Businesses are increasingly focused on data visualization due to the combination of increased data availability and easy-to-use software tools. In today’s workplace, employers expect accounting graduates to know how to use data visualization to understand, explain, and communicate answers to data questions.

There are two types of data visualization:

- **Exploratory data visualization** uses data visualization tools and techniques to explore data to find insights. Exploratory data visualization helps understand the data and identify underlying patterns, trends, or anomalies.
- **Explanatory data visualization** uses data visualization tools and techniques to communicate the results of an analysis. It is used to explain the results of an analysis, show relationships in the data, and communicate insights.

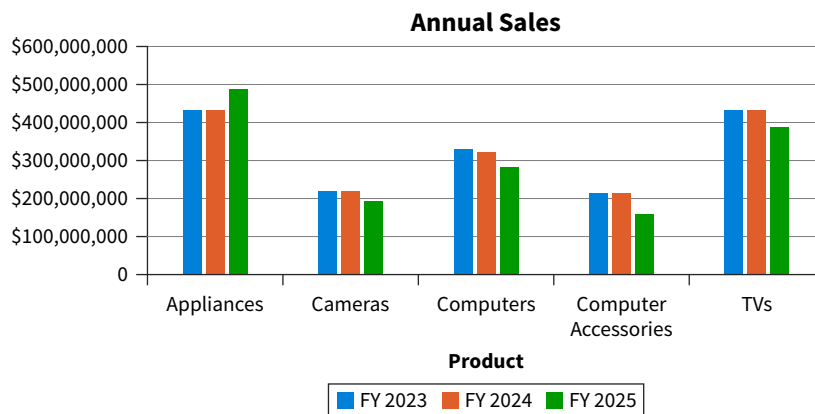
This course will eventually cover both in greater detail. This discussion introduces how visualizations make sense of large data sets, identifies common visualizations and when to use them, and explains how to create them in Microsoft Excel. Many accounting professionals prefer more powerful visualization software such as Power BI and Tableau because Microsoft Excel has limited data visualization capabilities. However, creating visualizations in Excel is an efficient and simple introduction.

Making Sense of Large Data Sets

Visualization is powerful because it can quickly and efficiently reveal insights buried in raw data. **Illustration 2.56** is a tabular summary of sales for an electronics store for the years 2023–2025. Compare the table to its visualization in column chart.

Annual Sales				
Product Category	FY 2023	FY 2024	FY 2025	Grand Total
Appliances	\$ 434,363,120	\$ 434,358,182	\$ 486,955,296	\$ 1,355,676,598
Cameras	\$ 217,276,947	\$ 217,491,906	\$ 192,648,723	\$ 627,417,576
Computers	\$ 327,140,700	\$ 323,513,274	\$ 281,322,005	\$ 931,975,979
Computer Accessories	\$ 216,412,502	\$ 215,524,232	\$ 157,419,883	\$ 589,356,617
TVs	\$ 432,403,786	\$ 434,187,852	\$ 389,195,034	\$ 1,255,786,672
Grand Total	\$ 1,627,597,055	\$ 1,625,075,446	\$ 1,507,540,941	\$ 4,760,213,442

ILLUSTRATION 2.56 Visualizing Sales Data



The graph of sales is easier to interpret:

- It is immediately clear that appliances and TVs have the highest sales in all three years, and only appliances had an increase in sales in 2025.
- All other product categories had lower sales in 2025 than the prior two years.

While it is possible to come to the same conclusions using the table, it is not as easy to detect the differences or make comparisons. The ability to quickly see patterns and relationships in large data sets are why data visualizations skills are so important.

Visualizations and When to Use Them

There are many types of visualizations available. Determining which to use is driven by the type of data available and what you are trying to show in the visualization. Data visualizations are covered in more detail in the analysis, interpretation, and communication chapters, but next is a summary of some common visualizations and how to select them.

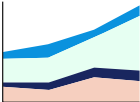
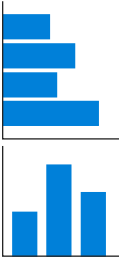
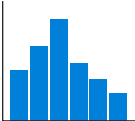
Common Visualizations

Categorical data are labeled or named data that can be sorted into groups according to specific characteristics. The data do not have a quantitative value. Categorical data are used in visualizations to portray groups of data. The visualization in Illustration 2.56 is an example using categorical data. The product types are the groups that summarize sales. The same visualization also includes quantitative data in the form of the sales amounts. Putting them together shows the relationship between product categories and sales. Notice the graph also has years. The bars represent the years 2023–2025.

Always consider whether the data being analyzed can be used in a particular visualization. For example, showing a relationship in the data using a scatterplot requires at least one quantitative measure. Showing trends over time requires a time measure (dates) plus a quantitative measure.

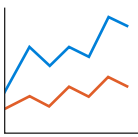
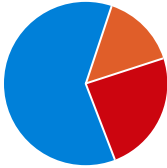
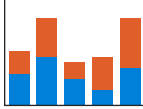
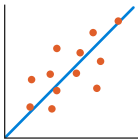
Illustration 2.57 lists some common data visualizations along with a description, best practices, and the types of data needed to create the visualization.

ILLUSTRATION 2.57 Common Data Visualizations

Visualization	Use	Best Practices	Data Required
Area Chart 	Representing changes in volume over time.	- Do not use if data has more than four categories to avoid confusion and clutter. - Start the y-axis at zero or above. - Put highly variable data on the top and data with low variability on the bottom.	- Date field. - At least one quantitative measure.
Bar and Column Chart 	Comparing parts to a whole, highlighting categories, or showing changes over time.	- Compare two to seven categories with vertical bars. - Use horizontal bars if there are more than seven categories or long category labels. - Use horizontal labels for readability. - Space bars should be used appropriately and consistently. - Use color sparingly or as an accent. - Always have a zero baseline.	- Horizontal bars: (bar chart) zero or more categories, one or more measures. - Vertical bars: (column chart) one or more categories, one or more quantitative measures.
Bubble Chart 	Comparing independent values that have distinct gaps or outliers.	- Label bubbles and make sure they are visible. - Scale bubble size by area and not diameter. - Do not use bubbles if they are all similar in size.	- One or more categories. - One or more quantitative measures.
Histogram Chart 	Showing frequency distributions.	- Use a zero baseline. - Choose an appropriate number of bins: - Bins are numbers that represent the intervals into which the data will be grouped. - Bins define the groups used for the frequency distribution. - Use between 5 and 15 bins.	Numerical data

(Continued)

ILLUSTRATION 2.57 (Continued)

Visualization	Use	Best Practices	Data Required
Line Graph 	Displaying one or more series of data. Allows for the use of multiple data series and data points.	• Time runs from left to right. • Be consistent plotting time points. • Use solid lines, not dotted. • Use a zero baseline. • Do not plot more than four lines. Use multiple charts instead.	• Continuous lines require one date, zero or more categories, and one or more quantitative measure.
Pie Chart 	Illustrating simple part-to-whole relationships. Not suited for making precise comparisons.	• Most impactful with small data sets. • Best for showing differences within groups based on one variable. • Make sure the data adds up to 100%. • Limit to a max of five segments. • Start first segment at 12 o'clock position.	• One or more categories. • One or two quantitative measures.
Stacked Bar Chart 	Comparing multiple part-to-whole relationships.	• Can be vertical or horizontal. • Follow same best practices as bar charts.	• One or more categories. • One or more quantitative measures.
Scatterplot (Scatter Chart) 	Highlighting correlation and distribution of large amounts of data.	• Data set should be in pairs with an independent variable (x-axis) and a dependent variable (y-axis). • Use if order is not relevant – otherwise use a line graph. • Do not use if there are only a few pieces of data or if there is no correlation.	• Zero or more categories. • One or more quantitative measures.
Tree Map 	Visualizing a part-to-whole relationship among many categories.	• Appropriate when precise comparisons are not important. • Use bright, contrasting colors so each box is easily defined. • Label boxes with text or numbers.	• One or more categories. • One or two quantitative measures.

As you develop your data analysis skills, you will be working with large sets of data and will need to use data visualization to analyze them and communicate findings. This chart can be a useful reference as you explore data and communicate your findings.

Choosing Visualizations

How do you know which visualization is best for an analysis? Start by considering the project's objective. There are some common analysis objectives:

- Showing composition
- Indicating relationships
- Displaying distributions
- Finding trends
- Making comparisons

Illustration 2.58 breaks down visualization choices if the goal is to show composition, relationships, or distributions.

ILLUSTRATION 2.58
Guide to Visualizations that Show Composition, Relationships, and Distributions

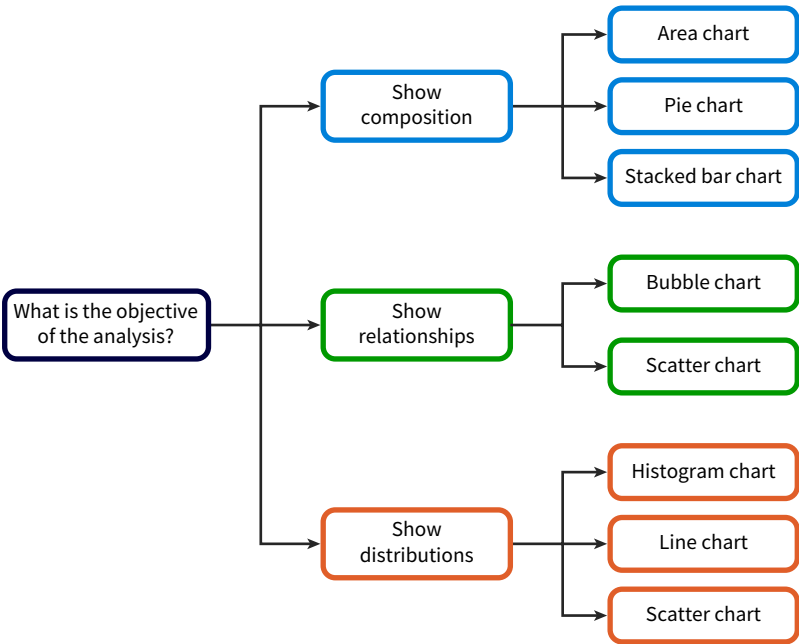
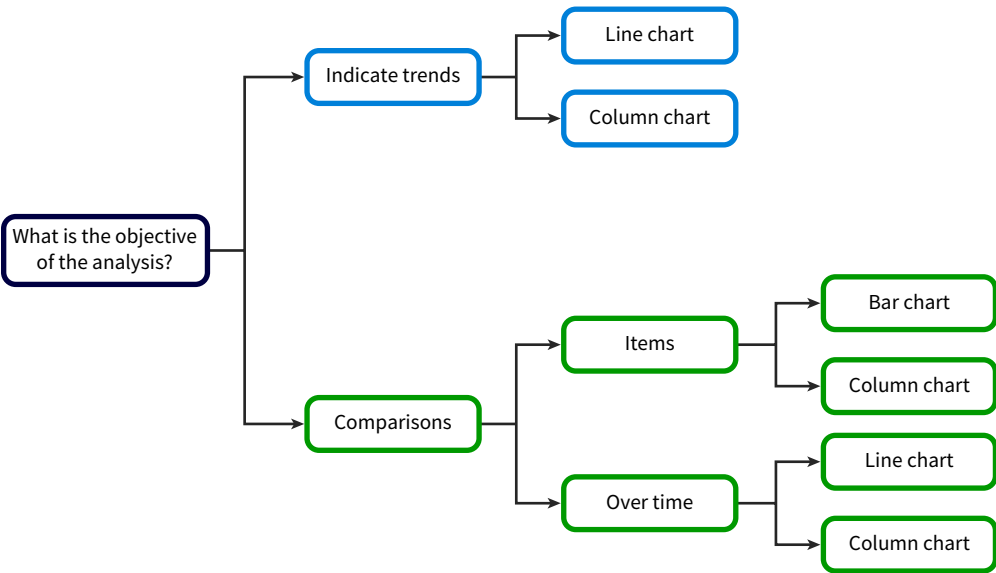


Illustration 2.59 identifies useful visualizations for showing trends or making comparisons.

ILLUSTRATION 2.59
Guide to Visualizations Showing Trends or Comparisons



You can use both visualization guides, along with the descriptions and best practices in Illustration 2.57, to create the best visualization to address the purpose of the analysis.

Microsoft Excel Visualizations

Many tools can create visualizations. This course focuses on a trio of software, Tableau, Power BI, and Microsoft Excel, which are the most common (but certainly not only) tools used in business today.

Microsoft Excel can create basic data visualizations. The visualization tools in Microsoft Excel are in the **Insert** ribbon ([Illustration 2.60](#))

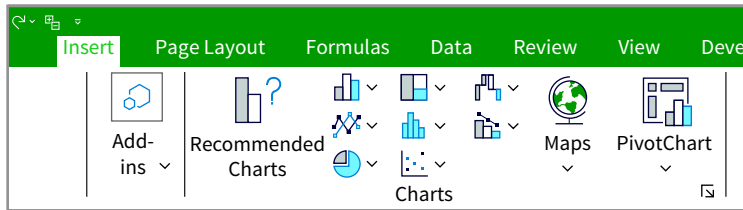


ILLUSTRATION 2.60 Insert a Visualization in Excel

To use the Charts tool:

- Highlight the data to chart.
- Choose a specific chart, or click on **Recommended Charts** to see suggestions.

A chart can also be created from an Excel PivotTable. Either create the PivotTable first or create the PivotTable and chart at the same time by using PivotChart. [Illustration 2.61](#) shows the dialog box when **Create PivotChart** is chosen. Notice that it is the same as the PivotTable dialog box.

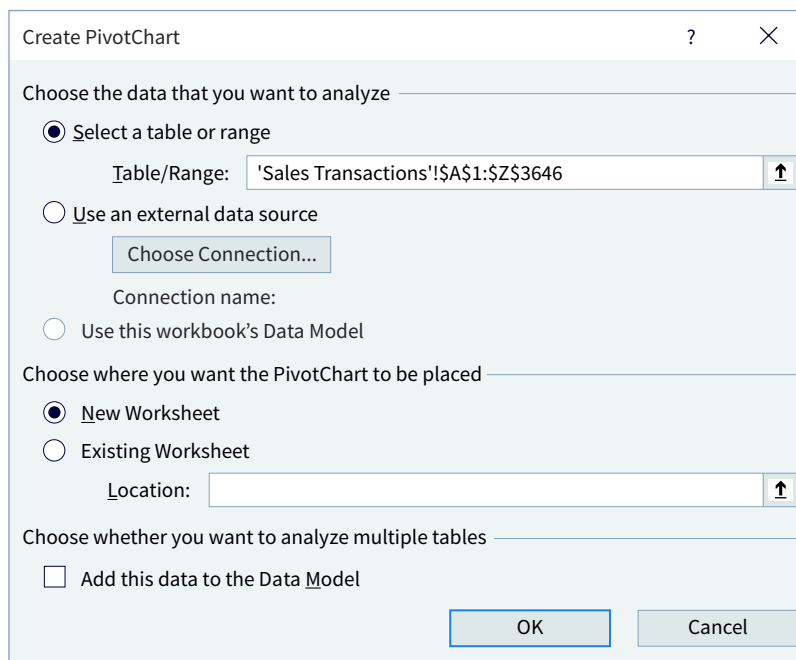


ILLUSTRATION 2.61 PivotChart Dialog Box

Data This example uses the Super Scooters data file to create a PivotTable and PivotChart of gross sales by year and model simultaneously ([Illustration 2.62](#)). To do this, select:

- The field **Model** for **Legend** (Series).
- The **Year** for **Axis** (Categories).
- **Sum of Gross Sales** for **Values**.

[Illustration 2.63](#) shows the resulting PivotTable and PivotChart, which is a bar chart in which the bar colors represent the models.

ILLUSTRATION 2.62 PivotTable and PivotChart Canvas

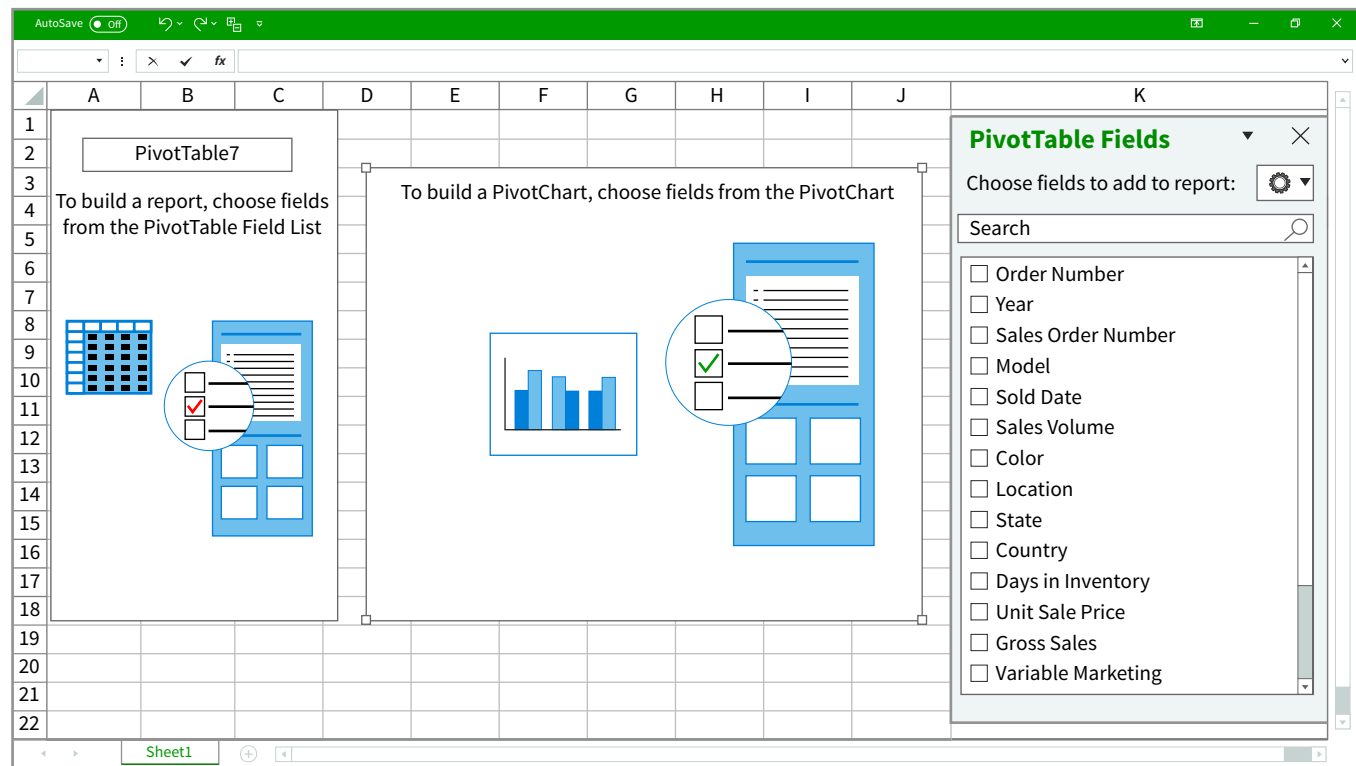
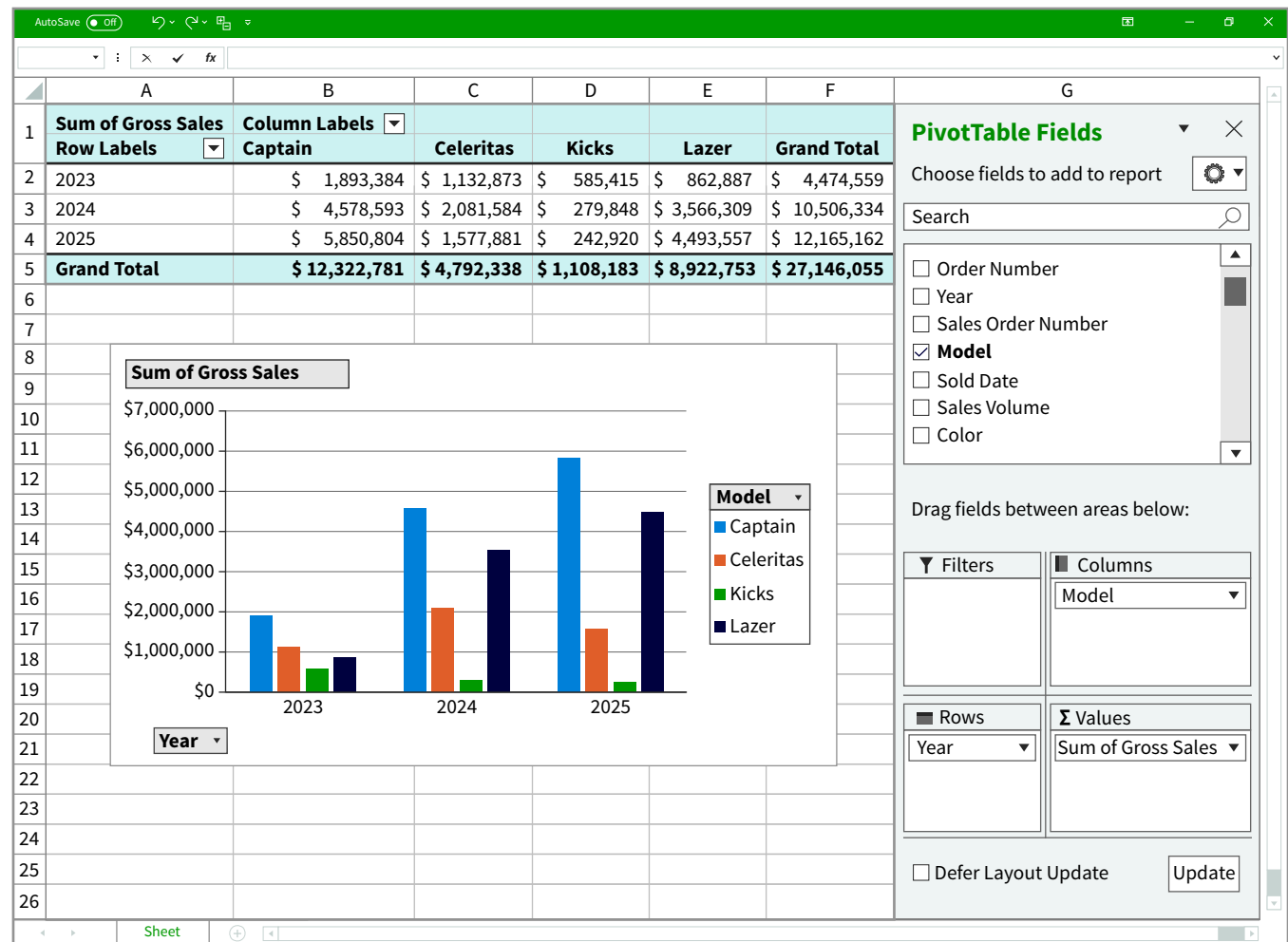


ILLUSTRATION 2.63 Super Scooters Completed PivotTable and PivotChart



During your accounting career you will encounter many types of data visualization software programs. However, learning how to use any of these software programs helps prepare you to use and learn new technology. (**Data** How To 2.2 is a walk-through for how to create a bar chart using Tableau.)



Managerial Accounting The controller at Super Scooters has asked you to prepare an analysis of product costs. Specifically, the controller would like answers to four questions:

1. What is the total cost per model each year?
2. Which variable costs are the highest?
3. Have labor, materials, and overhead costs been increasing or decreasing over time?
4. Are total costs related to sales volume?

For each question:

- Identify an appropriate visualization and explain your reasoning.
- List best practices for the visualization.
- Determine the type of data needed for the visualization.

Apply It 2.5

Analyze Product Costs with Data Visualization



SOLUTION

1. An area chart is the best choice because it represents changes in volume over time. For this question, it would show changes in total costs over time.

Best Practices:

- Do not use for data with more than four categories to avoid confusion and clutter. There are four models, so this requirement is met.
- Start the y-axis at zero or above.
- Place highly variable data on the top and data with low variability on the bottom.

Data:

- Date field (use year)
- At least one quantitative measure (use total costs)

2. A bar chart is the most appropriate visualization because it compares parts to a whole, highlights categories, or shows changes over time.

Best Practices:

- Compare two to seven categories with vertical bars.
- Use horizontal bars if there are more than seven categories or long category labels.
- Use horizontal labels for better readability.
- Space bars appropriately and consistently.
- Use color sparingly or as an accent.
- Always have a zero baseline.

Data:

- One or more categories (labor costs, materials, overhead)
- One or more quantitative measures (costs)

3. A line chart is appropriate because it displays one or more series of data and allows the use of multiple data series and data points.

Best Practices:

- Time runs from left to right.
- Be consistent plotting time points.
- Use solid lines, not dotted.
- Use a zero baseline.
- Do not plot more than four lines. Instead, use multiple charts.

Data:

- Continuous lines need one date, zero or more categories, and one or more quantitative measures.
- For this question there are years, three categories of expenses, and the costs of those expenses.

4. A scatterplot is the best choice because it highlights correlation and distribution of large amounts of data.

Best Practices:

- Data set should be in pairs with an independent variable (x-axis) and a dependent variable (y-axis).
- Use only if order is irrelevant, otherwise use a line graph.
- Avoid if there are only a few pieces of data or if there is no correlation.

Data:

- Zero or more categories
- One or more quantitative measures

Chapter Review and Practice

Learning Objectives Review

1 Describe how data is stored in and extracted from relational databases.

Data used in accounting data analyses is often stored in a relational database. Retrieving data from a database requires understanding how tables in a relational database are structured:

- Data is stored in tables, which are comprised of fields and records. Fields are the columns representing the characteristics about each record stored in the columns of the data set. Records are data in the rows representing instances of the phenomena being captured in the data set.
- Tables have a primary key, which is a unique value for each row in the table. Often, a table will have a foreign key. A foreign key is a primary key column repeated from another table. Foreign keys make it possible to join the data stored in different tables.
- To analyze data stored in more than one table, join the data from multiple tables. There are inner joins, right joins, left joins, or full joins, each of which retrieves data in different ways. The join selected should be appropriate for the data analysis question being asked.

2 Explain how functions help answer data analysis questions

Data analysis includes performing calculations:

- Functions are predefined formulas that perform frequently used calculations.

- The most common functions include AVERAGE, AVERAGEIF, AVERAGEIFS, COUNT, COUNTIF, COUNTIFS, SUM, SUMIF, and SUMIFS.

3 Illustrate how pivot tables organize and filter data.

Data organization is the process of rearranging data to make it easier to understand or to answer a specific question:

- Sort, filter, and slice are common tools for reorganizing data in a spreadsheet to answer questions.
- Pivot tables efficiently reorganize data in a spreadsheet to create custom summaries of key information.

4 Identify descriptive measures used to perform data analysis

The core data analysis skills for descriptive analytics are descriptive statistics and correlation analysis:

- Measures of location include the mean, median, and mode.
- Measures of distribution include variance and standard deviation.
- Measures of shape include skewness and kurtosis.
- Correlation measures can help identify relationships among the data. Correlation for continuous variables is measured using the correlation coefficient. This measure is a numerical value between -1 and $+1$. The closer the value is to the absolute value of 1 , the stronger the correlation.

5 Summarize how data visualization explores and explains data.

Data visualization is one of the fastest growing areas of data analysis in the accounting profession:

- Data visualization is the graphical presentation of data and information. Visualizing data can help quickly understand large sets of data.

- Exploratory data visualization examines data to uncover patterns, trends, or anomalies. Explanatory data visualization uses data visualization tools and techniques to communicate the results of data analysis.
- Choosing the right visualization is a combination of considering the purpose of the analysis and deciding whether the goal is to show composition, relationships, distributions, trends, or comparisons.

Key Terms Review

Attributes 2-2	Foreign key 2-3	Mode 2-26
Categorical data 2-42	Frequency distribution 2-30	Null value 2-4
Coefficient of kurtosis (CK) 2-30	Functions 2-8	Pivot table 2-13
Coefficient of skewness (CS) 2-30	Histogram 2-30	Primary key 2-3
Correlation analysis 2-34	Join 2-4	Query 2-4
Correlation coefficient 2-34	Kurtosis 2-30	Relational database 2-2
Data organization 2-13	Mean 2-26	Skewness 2-29
Slicing 2-22	Measures of central tendency 2-26	Slicers 2-22
Data visualization 2-40	Measures of dispersion 2-28	Standard deviation 2-28
Dimensions 2-12	Measures of location 2-26	Structured Query Language (SQL) 2-4
Explanatory data visualization 2-41	Measures of shape 2-29	Table 2-2
Exploratory data visualization 2-41	Median 2-26	Variance 2-28

How To Walk-Throughs

HOW TO 2.1 Format and Show Values As Options in PivotTables

Values in an Excel PivotTable can be formatted with the Value Field Settings dialog box.

What You Need: **Data** The How To 2.1 data file.

STEP 1: Click **Number Format**. The same **Format Cells** dialog available in the Home tool ribbon will appear (**Illustration 2.64**).

How To

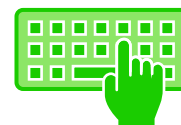
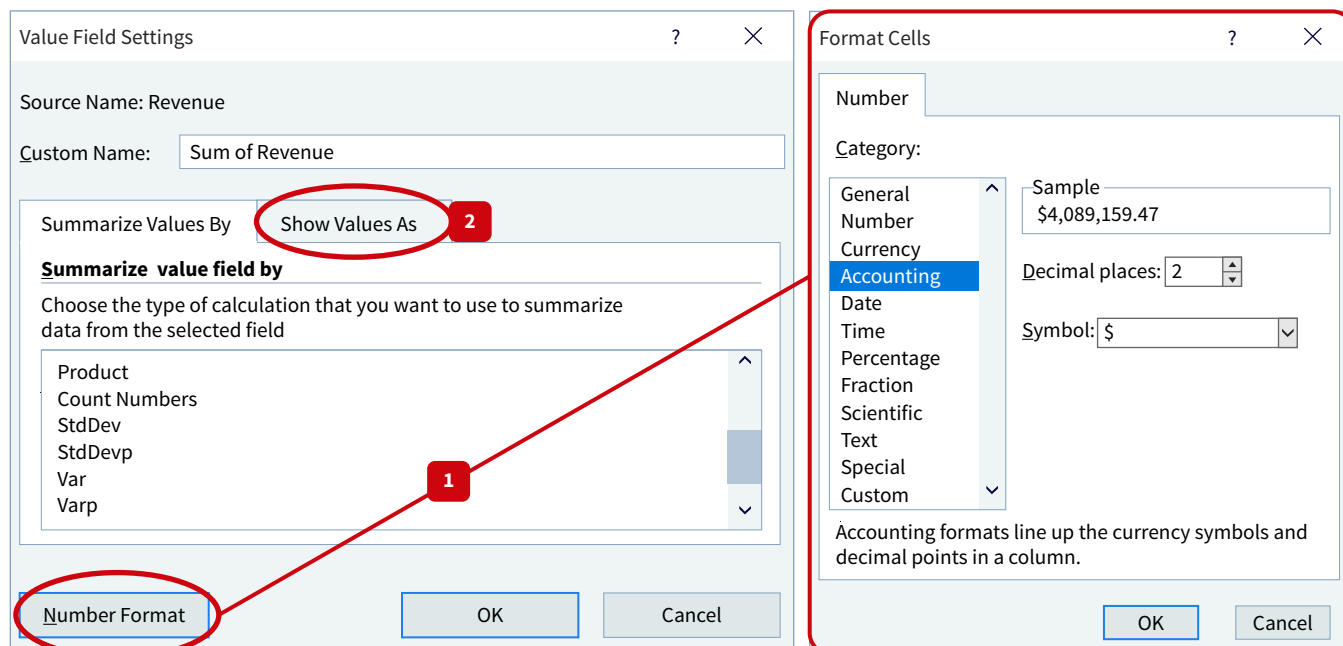


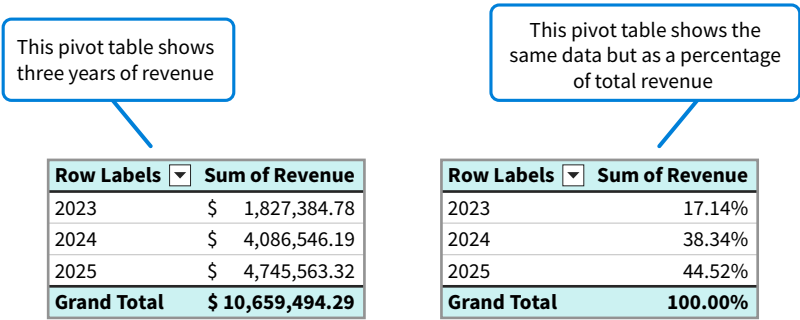
ILLUSTRATION 2.64 Format Values in a PivotTable



STEP 2: It is also possible to use the **Show Values As** option in the **Value Field Settings** dialog box to add a quick calculation of the values:

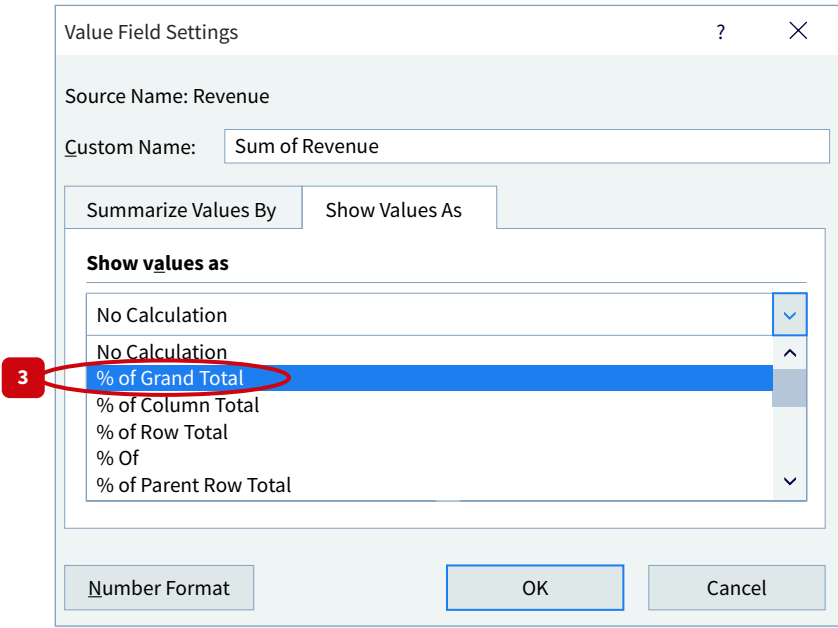
- Clicking on **Show Values As** reveals a dropdown list of built-in calculations.
- **Illustration 2.65** displays values as a percentage of a grand total.

ILLUSTRATION 2.65 Show Values as Options



STEP 3: The **Sum of Revenue** can quickly be changed to the **Percent of Total Revenue** by selecting **% of Grand Total** in the dialog box (**Illustration 2.66**).

ILLUSTRATION 2.66 Show Values As Options



HOW TO 2.2 Create a Bar Chart Using Tableau

The same visualization from Illustration 2.63 can be created using Tableau.

What You Need: **Data** The How To 2.2 data file.

STEP 1: Add the fields for the visualization to the canvas:

- Open the file and click **Sheet 1** along the bottom of the screen ([Illustration 2.67](#)).
- This will open a new worksheet for a visualization. Click on the field to visualize and drag it to the column or row line:
 - Drag **Sold Date** to **Columns**, **Model** to **Rows**, and **Gross Sales** to **Text**.
 - You could also drag it to the desired spot in the canvas in the area labeled **Drop field here**.

ILLUSTRATION 2.67 Tableau Visualization Canvas

The screenshot shows the Tableau interface for a file named "Tableau - Super Scooters How To 2.2". The main pane displays the "Data" section with a list of fields categorized into "Dimensions" and "Measures".

Dimensions: Color, Country, State, Location, Model, Order Number, Sales Order Number, Sold Date, Year.

Measures: Contribution Margin, Days in Inventory, Gross Sales, Labor, Materials, Overhead, Revenue, Sales Tax, Sales Tax Rate, Sales Volume, Total Allocated Fixed Costs, Total Costs, Total Depreciation, Total Fixed Costs, Total Variable Costs, Total Warranty, Unit Sale Price, Variable Marketing, Latitude (generated), Longitude (generated), Sales Transactions (Count), Measure Values.

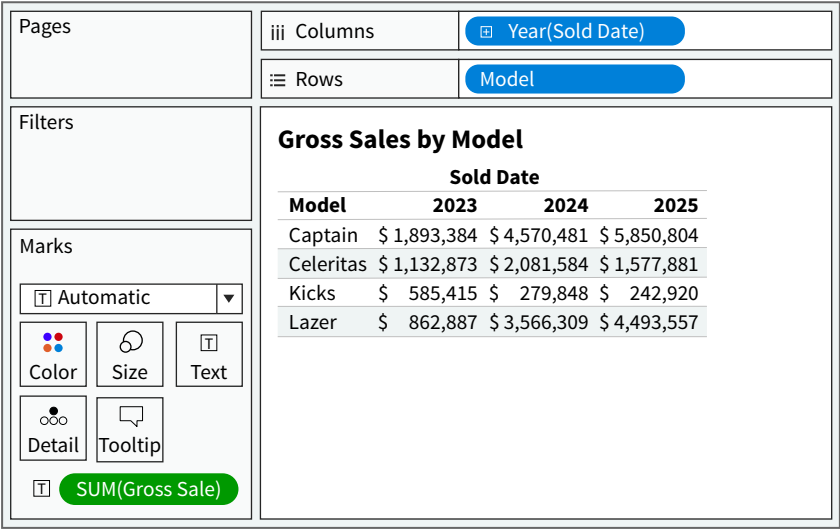
The "Marks" card is set to "Automatic" and includes fields for Color, Size, Text, Detail, and Tooltip. The "Columns" and "Rows" shelves are empty. The "Drop field here" area is highlighted in the bottom right.

Annotations:

- A blue box points to the "Tables" list, stating: "The list of tables. In this file there is only one."
- A blue box points to the "Dimensions" section, stating: "All the fields above the line are considered dimensions. All the fields below the line are measures. Measures are continuous numeric fields. Dimensions are fields that are not measures."
- Red dashed lines and circles highlight the fields "Model", "Sold Date", and "Gross Sales" being dragged to the "Columns", "Rows", and "Text" shelves respectively.

Illustration 2.68 shows the result when **Year** is dragged to **Columns**, **Model** to **Rows**, and **Gross Sales** to **Text**.

ILLUSTRATION 2.68 Tableau Visualization Canvas: Step 1



STEP 2: Create a bar chart. Notice Tableau defaulted to a Table format.

- Click **Show Me** at the top right corner of the screen.
- Choose the side-by-side column chart. Note that any of the highlighted visualizations can be chosen (**Illustration 2.69**).

ILLUSTRATION 2.69 Tableau Show Me Options

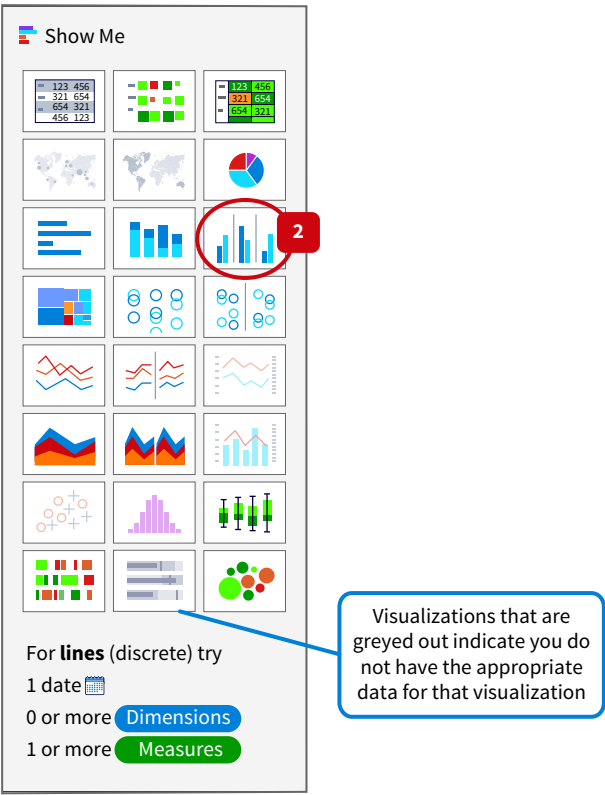


Illustration 2.70 is the result of selecting the column bar chart visualization option.

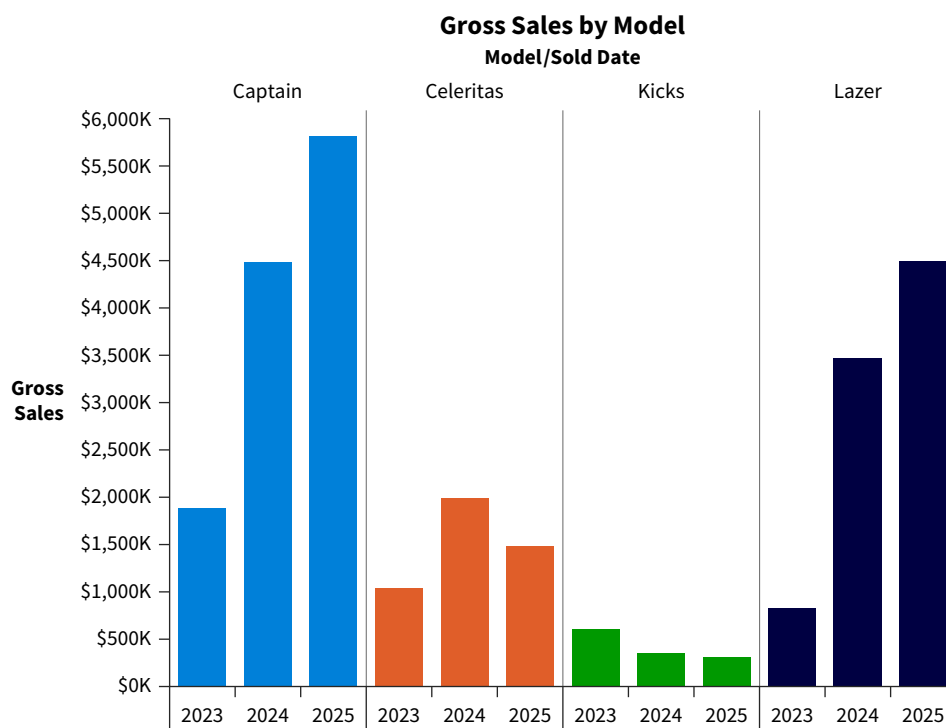


ILLUSTRATION 2.70 Tableau Bar Chart Visualization

STEP 3: Title the visualization:

- Double-click on the **Sheet 1** title and choose **Edit Title**.
- Change the title by typing “Gross Sales by Model” (**Illustration 2.71**).

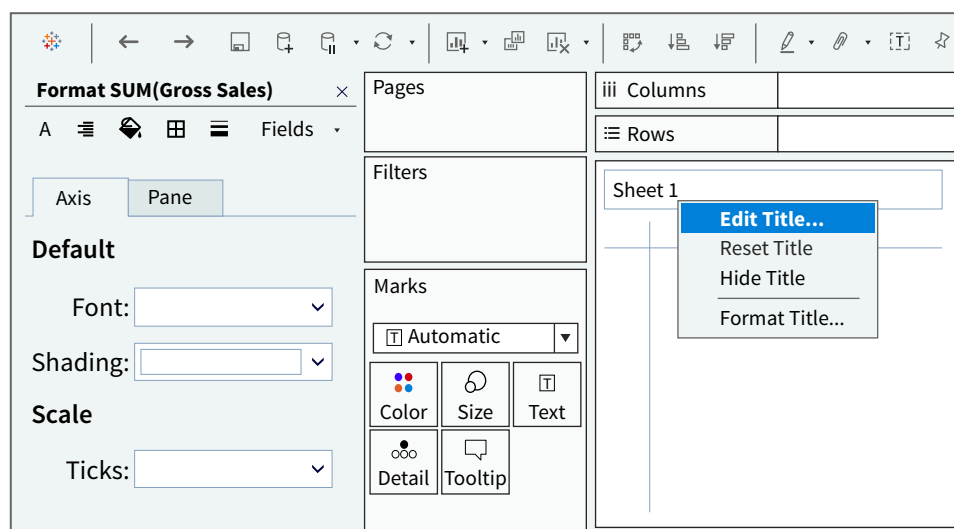
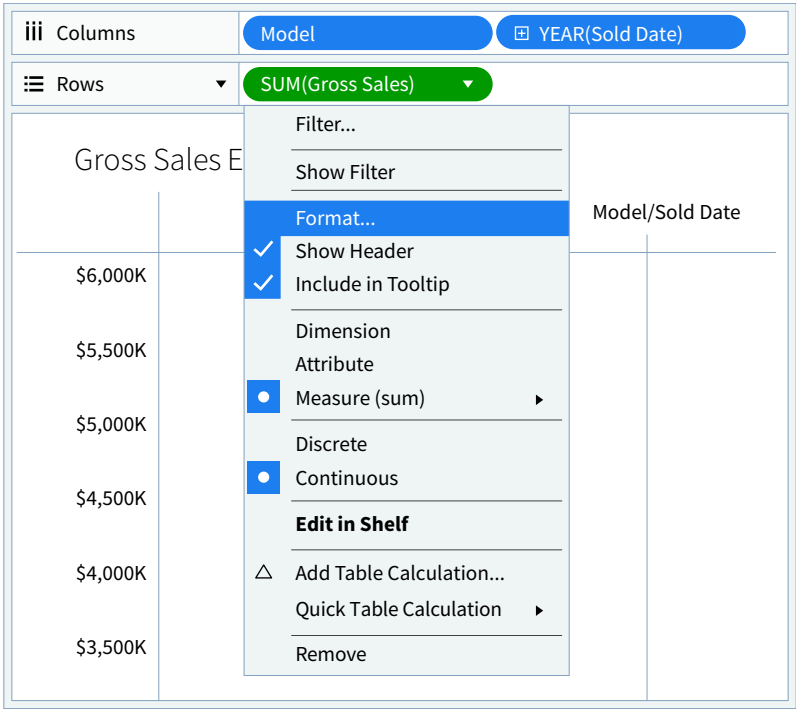


ILLUSTRATION 2.71 Tableau Bar Chart: Add a Title

STEP 4: Format the axis:

- To format the Gross sales axis and dollars in thousands, click the down arrow in the **SUM** (Gross Sales).
- Choose **Format** and click the **Axis Tab** (**Illustration 2.72**).

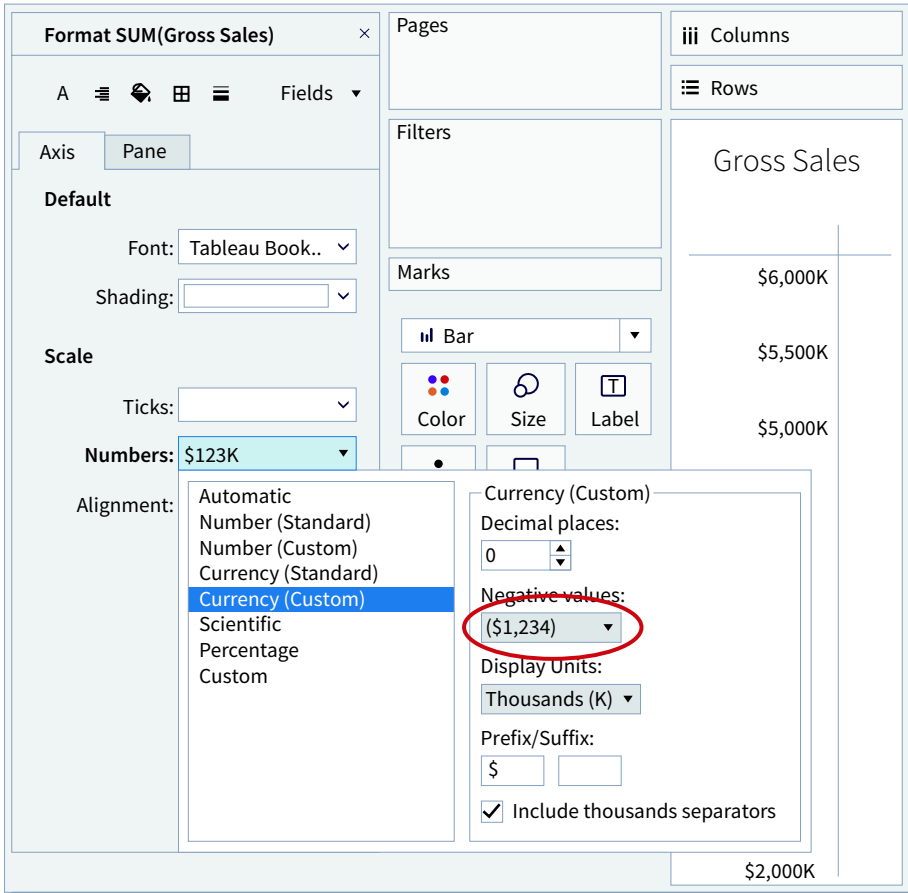
ILLUSTRATION 2.72 Tableau Bar Chart: Format the Axis



STEP 5: Change currency display to thousands:

- Once in the **Format** box, choose **Numbers** and **Currency (Custom)**.
- Select **Display Units** to change the numbers to be shown in thousands.
- Note, we have also changed the decimal places to zero ([Illustration 2.73](#)).

ILLUSTRATION 2.73 Tableau Bar Chart: Change Currency Display



Data The Data tag appears when the data required to answer a question or complete an exercise are available on Wiley's online learning platform.

Multiple Choice Questions

1. **(LO 1)** A collection of logically related data that can be retrieved, manipulated, and updated to meet users' needs is called a
 - a. table.
 - b. relational database.
 - c. data set.
 - d. data warehouse.
2. **(LO 1)** In a relational database table, a primary key
 - a. is not always needed.
 - b. is the same as a foreign key.
 - c. is a unique value.
 - d. can be repeated in the table if needed.
3. **(LO 1)** Tables in a database may be in which of the following elements?
 - a. Assets, liabilities, expenses
 - b. Resources, expenses, employees
 - c. Revenue, events, agents
 - d. Resources, events, agents
4. **(LO 1)** Columns in a database reflect
 - a. attributes.
 - b. events.
 - c. resources.
 - d. agents.
5. **(LO 1)** A request for data from a database in order to retrieve or manipulate it is called
 - a. data analysis.
 - b. a query.
 - c. a question.
 - d. a join.
6. **(LO 1)** When performing a join between two tables, if a null value is returned, that indicates
 - a. a value of zero.
 - b. the join was incorrect.
 - c. a value does not exist in the database.
 - d. an inner join was performed.
7. **(LO 1)** A join that results in retrieving all the records from the left table and the matching records from the right table is which type of join?
 - a. Inner
 - b. Left
 - c. Right
 - d. Full
8. **(LO 1)** A join that returns all rows from both tables that have matching values is which type of join?
 - a. Inner
 - b. Left
 - c. Right
 - d. Full
9. **(LO 1)** A join that returns all the records from the right table as well as all the matched records from the left table is which type of join?
 - a. Inner
 - b. Left
 - c. Right
 - d. Full
10. **(LO 1)** A join that returns all records when there is a match in either the left or right table is which type of join?
 - a. Inner
 - b. Left
 - c. Right
 - d. Full
11. **(LO 2)** Which of the following Microsoft Excel functions returns the arithmetic mean for a range or array of numbers?
 - a. AVERAGE
 - b. MEAN
 - c. AVERAGEIF
 - d. SUM
12. **(LO 2)** Which of the following Microsoft Excel functions counts the number of cells specified by a given set of criteria?
 - a. COUNT
 - b. COUNTIF
 - c. COUNTIFS
 - d. COUNTA
13. **(LO 2)** This function adds the cells specified by a given set of conditions or criteria.
 - a. SUM
 - b. SUMIF
 - c. SUMIFS
 - d. SUMPRODUCT
14. **(LO 3)** This function adds the cells specified by a condition or criteria.
 - a. SUM
 - b. SUMIF
 - c. SUMIFS
 - d. SUMPRODUCT
15. **(LO 3)** Which tool in Microsoft Excel will reorganize spreadsheet data into custom summaries of key information?
 - a. Sort
 - b. Filter
 - c. PivotTable
 - d. Developer
16. **(LO 3)** A PivotTable displays sales by product type, year of sale, and the total dollar amount of sale. You would like to only see three of the 15 specific product types. The best way to focus only on the specific product types you are interested in is to use
 - a. the SORT function.
 - b. the SUMIF function.
 - c. a filter.
 - d. cut and paste.
17. **(LO 3)** Auto filter functionality in a PivotTable can be accessed by
 - a. using the filter box in the PivotTable fields.
 - b. clicking on the dropdown arrow in the row labels.
 - c. clicking on the dropdown arrow in the filter box.
 - d. dragging a field to the field filter box.
18. **(LO 3)** Slicing refers to
 - a. removing decimal places.
 - b. eliminating data.
 - c. sorting data.
 - d. breaking down the data into smaller parts.

19. (LO 4) The sum of all observations in a data set divided by the total number of observations is called the

- a. mean.
- b. median.
- c. mode.
- d. range.

20. (LO 4) The middle value when the data in a data set are arranged from smallest to largest is called the

- a. mean.
- b. median.
- c. mode.
- d. range.

21. (LO 4) Measures of dispersion show

- a. central tendency.
- b. shape.
- c. variation.
- d. location.

22. (LO 4) If the coefficient of skewness is -1.5, the data distribution will have a

- a. high degree of skewness and tail to the right.
- b. high degree of skewness and tail to the left.
- c. moderate degree of skewness and tail the right.
- d. moderate degree of skewness and tail to the left.

23. (LO 4) A negative correlation means

- a. as one variable decreases, the other variable decreases.
- b. as one variable increases, the other variable increases.
- c. as one variable increases, the other variable decreases.
- d. as one variable decreases, the other variable remains unchanged.

24. (LO 5) Which of the following are benefits of data visualization?

- a. Visualizing helps to quickly understand large data sets.
- b. Data visualization can be used to explore data.
- c. Data visualization can be used to explain data analysis.
- d. All of these are benefits of data visualization.

25. (LO 5) Using data visualization to identify underlying patterns is considered

- a. explanatory data visualization.
- b. exploratory data visualization.
- c. graphical analysis.
- d. top-down analysis.

26. (LO 5) A visualization used to represent changes in volume over time is a/an

- a. area chart.
- b. bar chart.
- c. line graph.
- d. pie chart.

27. (LO 5) A visualization used to illustrate simple part-to-whole relationships is a/an

- a. area chart.
- b. bar chart.
- c. line graph.
- d. pie chart.

28. (LO 5) A visualization used to show frequency distributions is a/an

- a. pie chart.
- b. line chart.
- c. histogram chart.
- d. bubble chart.

Review Questions

1. (LO 1) Identify and describe the four types of joins that can query a data set.

2. (LO 2) For each scenario, match the best basic Excel function that could be used to address it. Each function can be used once, more than once, or not at all.

- a. AVERAGE
- b. AVERAGEIF
- c. COUNT
- d. COUNTIF
- e. COUNTA
- f. COUNTBLANK
- g. SUM
- h. SUMIF

Scenario	Function
1. Count the number of cells in an Excel file that have inventory quantities.	
2. Count the number of cells in an Excel file that have an inventory quantity of 1,150 items.	
3. Calculate the arithmetic mean of the commission amounts paid to sales personnel during the fourth quarter.	

Scenario	Function
4. Calculate the total sales amount for the period, which is listed in column K of your Excel spreadsheet.	
5. Calculate the total sales amount for the period for customer #4920 only. The sales amounts are listed in column K in your Excel spreadsheet and customer numbers are listed in column A of your Excel spreadsheet.	
6. Count the number of inventory items listed on the spreadsheet with no inventory quantities.	

3. (LO 2) Give an example of when you should use a COUNTIFS function.

4. (LO 2) Give an example of when you might use a COUNTBLANK function.

5. (LO 3) Describe the five components of an Excel PivotTable.

6. (LO 3) Describe how an Excel PivotTable can be filtered.

7. (LO 4) Provide an example of when the median of a distribution might be more meaningful to interpret than the mean.

8. (LO 4) Define standard deviation and give an example of how to interpret it.

9. **(LO 4)** Define negative correlation and positive correlation. Provide an example of each.
10. **(LO 5)** Describe exploratory data visualization and explanatory data visualization. How are they the same? How are they different?
11. **(LO 5)** For each scenario identify whether you would perform exploratory data visualization or explanatory data visualization.

Scenario	Visualization Type
1. Your manager provides you with all the sales data by product line for the last two years and asks you to identify sales trends between years.	
2. You have analyzed data related to sales trends by country over the past three years and will present that data using a tree map.	
3. Your manager provides you with all the payments made to approved vendors in the past six months and asks you to identify if any payments are outside of expected payment amounts.	
4. Your manager provides you with analysis of maintenance expenses for the year and asks you to prepare a pie chart to illustrate the categories of expenses.	

12. **(LO 5)** As a tax professional for an online retailer, you have been asked to create a line graph visualization depicting the increase in sales tax collected and remitted each month this year compared to last year. Using the line graph best practices outlined in the chapter, describe how you would set up the visualization. Identify the data points on the x-axis, y-axis, and series of data.

13. **(LO 5)** Describe the best practices for bar charts and area charts.

14. **(LO 5)** You are a financial analyst in the operations division of a manufacturing company. As part of quality control metrics, your company tracks rework expenses for goods that were not manufactured to meet quality standards. Your manager has asked you to prepare a line chart illustrating the trend of rework expenses by reason code category. You note that there are five different reason codes for why rework may be required. Using the best practices outlined in this chapter, describe how you would set up the line chart to depict rework expense trends by category.

Brief Exercises

BE 2.1 (LO 1) Managerial Accounting You are a financial analyst for PizzaNow! The corporate controller wants you to perform an analysis using three tables in the relational database.

For each item, identify whether it is a primary key, foreign key, or neither.

TakeOrder	Customers	Employees
OrderNumber CustomerID EmployeeID Date Time Notes	CustomerID LastName FirstName StreetAddress City State ZipCode PhoneNumber	EmployeeID LastName FirstName DateHired DateOfBirth PayRate(hourly) JobTitle StreetAddress City State ZipCode PhoneNumber

- OrderNumber in the TakeOrder table
- EmployeeID in the TakeOrder table
- CustomerID in the Customers table
- EmployeeID in the Employees table
- Date in the TakeOrder table
- ZipCode in the Employees table

BE 2.2 (LO 1) Accounting Information Systems Dine At Home provides home delivery of meals ordered from various local restaurants. You are the liaison between the company's information technology group and the accounting group. You are asked to explain the relationship among these tables to the accounting group. The tables are from the Dine At Home database.

Restaurant	Order	Customer1
RestaurantNumber RestaurantName Street City State Zip OwnerFirstName OwnerLastName Phone Website	OrderNumber RestaurantNumber OrderDate OrderAmount CustomerID	CustomerID FirstName LastName Phone Site Address City State ZipCode

1. Identify the primary and foreign keys for each table.
2. If you want to know the name of a customer for a specific order, which tables should you query?

BE 2.3 (LO 1) Financial Accounting Assume you are a financial analyst in the controller's group for your distribution company. You have been asked to identify all inventory items without sales in the past year:

- The IT group has provided the inventory on hand data file and the twelve month sales data file.
- You have identified the inventory on hand table as the left table and the twelve month sales table as the right table.

Identify the most appropriate join for these two tables to perform your analysis. Why is this join the most appropriate?

BE 2.4 (LO 1) Financial Accounting You are a financial analyst for Dine At Home who has been asked to analyze the data in the following three tables.

Restaurant	Order	Customer1
RestaurantNumber RestaurantName Street City State Zip OwnerFirstName OwnerLastName Phone Website	OrderNumber RestaurantNumber OrderDate OrderAmount CustomerID	CustomerID FirstName LastName Phone Site Address City State ZipCode

For each scenario, identify the join you would most likely use to query the data. Each join type can be used once, more than once, or not at all.

- a. Left join
 - b. Right join
 - c. Inner join
 - d. Full join
1. Execute a query to join the Restaurant table (left table) and the Order table (right table), but only return the rows from both tables that have matching values.
 2. Execute a query to join the Restaurant table (left table) and the Customer1 (right table) table, and return all records from the Restaurant table, but only those matching records from the Customer1 table.
 3. Execute a query to join the Order table (left table) and the Customer1 table (right table), and return all records from both tables. Match the records that can be matched in both tables.

- Execute a query to join the Order table (left table) and the Customer1 table (right table), and only return all the records from the Order table and the matching records from the Customer1 table.

BE 2.5 (LO 2) Data Managerial Accounting The controller at ThisBigCity has asked that you perform an analysis on the city's employee reimbursement expenses over the past fifteen years. The IT group has provided a download of all the employee reimbursement data since 2005.

- Use the AVERAGE function. What was the average reimbursement amount paid from July 2005 to November 2020?
- Use the AVERAGEIF function. What was the average reimbursement amount paid in 2019?
- Use the AVERAGEIFS function. What was the average reimbursement amount paid in the fire department during 2019?

BE 2.6 (LO 2) Data Accounting Information Systems As an internal auditor at ThisBigCity, you testing internal controls over the employee reimbursement process for the city. The IT group provided a download of all the employee reimbursement data since 2005. Your manager suggested performing descriptive statistics on this file to determine if you have a complete population of data, and to start the process of identifying a sample size for internal control testing.

- Use the COUNT function. How many reimbursements were paid from July 2005 to November 2020?
- Use the COUNTIF function. How many reimbursements were paid in 2019?
- Use the COUNTIFS function. How many reimbursements were paid in 2019 to firefighters?

BE 2.7 (LO 3) Data Auditing Use PivotTables and the available data to answer the following questions:

- Which customer has the highest accounts receivable balance?
- Which customer has the highest accounts receivable balance that is more than 150 days past due?

BE 2.8 (LO 3) Data Financial Accounting Use PivotTables and the available data to answer the following questions:

- What is the total of accounts receivable?
- What are the totals by region?

BE 2.9 (LO 3) Data Financial Accounting Managerial Accounting Use PivotTables and the Super Scooters data to answer the following questions:

- What are the total gross sales for each scooter model by year?
- Which color scooter had the highest sales volume in 2023?
- What was the total variable marketing expense for 2023 by model?

BE 2.10 (LO 4) Data Managerial Accounting As a financial analyst working for Animal Control Centers, you want to understand the overtime pay during 2025. Find the following statistics for overtime pay:

- Mean
- Median
- Mode

BE 2.11 (LO 4) Data Financial Accounting You are preparing the management's discussion and analysis (MD&A) regarding the city of Chicago's fire department. One of the most important expenses for the fire department is overtime. Therefore, you would like to understand the overtime data before you write the MD&A.

- Calculate the coefficient of skewness for overtime pay.
- Calculate the coefficient of kurtosis for overtime pay.
- Prepare a histogram with these groupings: \$500, \$1,000, \$2,000, \$3,000, \$4,000, \$5,000, \$6,000.

BE 2.12 (LO 5) Data Financial Accounting Your company, Loans Are US, provides loans to small to medium-sized businesses. The company has loan offices in four regions. You are asked to prepare a visualization that illustrates the total amount of loans by region and by the age of receivables. Prepare a stacked column chart that does this.

BE 2.13 (LO 5) Data Loans Are US keeps track of credit ratings for all customer accounts. You must prepare a visualization that illustrates the total number of loans by credit rating. Prepare a bar chart visualizing the number of accounts in each of the three credit ratings: AAA, BBB, and CCC.

BE 2.14 (LO 5) Data Your supervisor at Loans Are US has asked you to prepare a visualization that illustrates the total dollar amount of loans past due by more than 150 days by credit rating. Prepare a bar chart visualizing the number of accounts in each of the three credit ratings: AAA, BBB, and CCC.

Exercises

EX 2.1 (LO 1) Auditing Identify Data and Joins Needed to Verify Data You are an internal auditor for Way Cool Stuff. You must verify that no employees are also customers. The following tables are in the database.

<table><tr><th>Locations</th></tr><tr><td> LocationNumber LocationDescription</td></tr></table>	Locations	 LocationNumber LocationDescription	<table><tr><th>SalesOrders</th></tr><tr><td> SalesOrderNumber OrderNumber Model SoldDate SalesVolume Color Location Region State Country UnitSalePrice CustomerNumber EmployeeNumber</td></tr></table>	SalesOrders	 SalesOrderNumber OrderNumber Model SoldDate SalesVolume Color Location Region State Country UnitSalePrice CustomerNumber EmployeeNumber	<table><tr><th>Employee</th></tr><tr><td> EmployeeNumber FirstName LastName Address City State ZipCode RegionNumber</td></tr></table>	Employee	 EmployeeNumber FirstName LastName Address City State ZipCode RegionNumber
Locations								
 LocationNumber LocationDescription								
SalesOrders								
 SalesOrderNumber OrderNumber Model SoldDate SalesVolume Color Location Region State Country UnitSalePrice CustomerNumber EmployeeNumber								
Employee								
 EmployeeNumber FirstName LastName Address City State ZipCode RegionNumber								
<table><tr><th>Regions</th></tr><tr><td> RegionNumber RegionDescription</td></tr></table>	Regions	 RegionNumber RegionDescription						
Regions								
 RegionNumber RegionDescription								
<table><tr><th>Inventory</th></tr><tr><td> ItemNumber ItemDescription Color QuantityOnHand</td></tr></table>	Inventory	 ItemNumber ItemDescription Color QuantityOnHand	<table><tr><th>SalesOrderExpenses</th></tr><tr><td> OrderNumber VariableMarketing Labor Materials Overhead TotalWarranty TotalDepreciation SalesTax</td></tr></table>	SalesOrderExpenses	 OrderNumber VariableMarketing Labor Materials Overhead TotalWarranty TotalDepreciation SalesTax	<table><tr><th>Customer</th></tr><tr><td> CustomerNumber CustomerName CustomerAddress CustomerCity CustomerState CustomerZipCode ContactFirstName ContactLastName PhoneNumber</td></tr></table>	Customer	 CustomerNumber CustomerName CustomerAddress CustomerCity CustomerState CustomerZipCode ContactFirstName ContactLastName PhoneNumber
Inventory								
 ItemNumber ItemDescription Color QuantityOnHand								
SalesOrderExpenses								
 OrderNumber VariableMarketing Labor Materials Overhead TotalWarranty TotalDepreciation SalesTax								
Customer								
 CustomerNumber CustomerName CustomerAddress CustomerCity CustomerState CustomerZipCode ContactFirstName ContactLastName PhoneNumber								

1. What table(s) will you need to be able to complete this test?
2. What fields will you use to join the tables, if necessary?
3. What fields do you need from the table(s) to complete this test?

EX 2.2 (LO 1) Financial Accounting Identify Data and Joins Needed to Summarize Data You are a financial accountant at Way Cool Stuff calculating net income. What follows are the database tables available for Way Cool Stuff.

Locations

LocationNumber

LocationDescription

Regions

RegionNumber

RegionDescription

Inventory

ItemNumber

ItemDescription

Color

QuantityOnHand

SalesOrders

SalesOrderNumber

OrderNumber

Model

SoldDate

SalesVolume

Color

Location

Region

State

Country

UnitSalePrice

CustomerNumber

EmployeeNumber

SalesOrderExpenses

OrderNumber

VariableMarketing

Labor

Materials

Overhead

TotalWarranty

TotalDepreciation

SalesTax

Employee

EmployeeNumber

FirstName

LastName

Address

City

State

ZipCode

RegionNumber

Customer

CustomerNumber

CustomerName

CustomerAddress

CustomerCity

CustomerState

CustomerZipCode

ContactFirstName

ContactLastName

PhoneNumber

1. Which table(s) do you need to be able to gather the data needed to calculate net income?
2. What fields will you use to join the tables?
3. Which fields will you need to calculate net income?

EX 2.3 (LO 1) Managerial Accounting Apply Joins to Answer Questions You are analyzing sales at Way Cool Stuff by region for the years 2024 and 2025. The following database tables are available.

Locations

LocationNumber
LocationDescription

Regions

RegionNumber
RegionName

Inventory

ItemNumber
ItemDescription
Color
QuantityOnHand

SalesOrders

SalesOrderNumber
OrderNumber
Model
SoldDate
SalesVolume
Color
Location
RegionNumber
State
Country
UnitSalePrice
CustomerNumber
EmployeeNumber

SalesOrderExpenses

OrderNumber
VariableMarketing
Labor
Materials
Overhead
TotalWarranty
TotalDepreciation
SalesTax

Employee

EmployeeNumber
FirstName
LastName
Address
City
State
ZipCode
RegionNumber

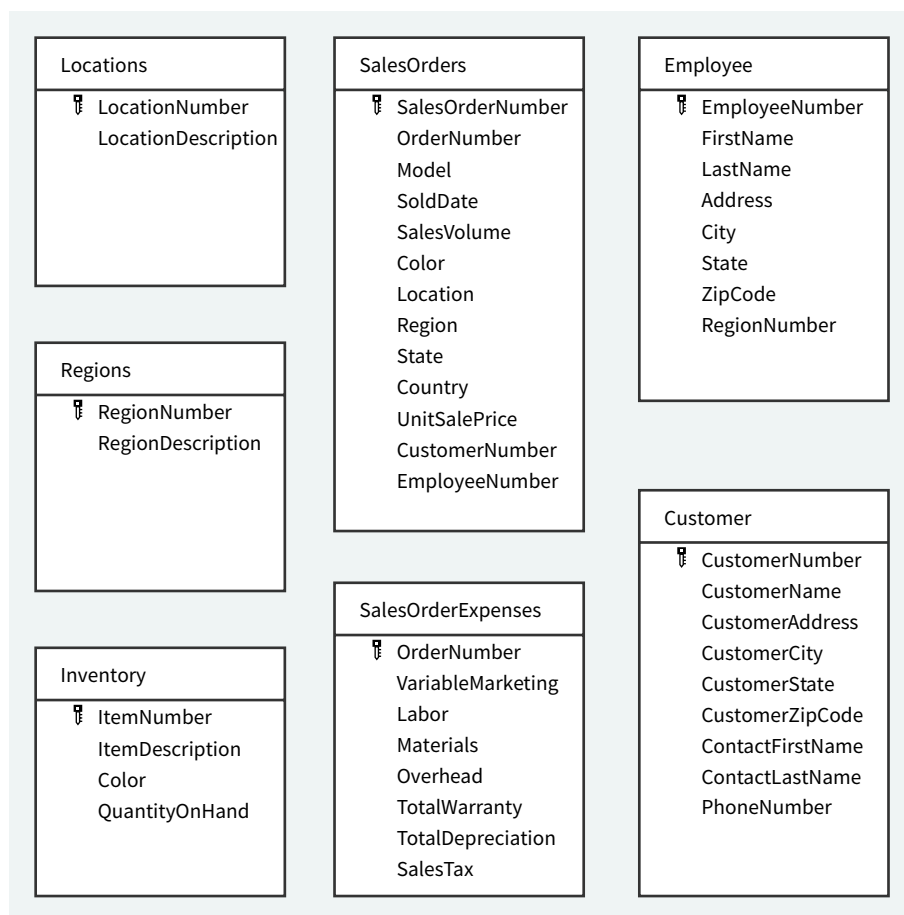
Customer

CustomerNumber
CustomerName
CustomerAddress
CustomerCity
CustomerState
CustomerZipCode
ContactFirstName
ContactLastName
PhoneNumber

1. Which table(s) do you need to gather the data necessary to analyze gross sales by region for 2024 and 2025?
2. What fields will you use to join the tables?
3. Which fields will you need to perform your analysis?

EX 2.4 (LO 1) Tax Accounting Identify Data and Joins Needed for Tax Compliance You are filing Way Cool Stuff's state sales tax returns for the month end December 31, 2025 for all locations in which sales tax was collected. Use the available database tables to answer the questions.

1. Which table(s) do you need to be able to gather the data needed to calculate sales tax remittance for December 2025?
2. What fields will you use to join the tables?
3. Which fields will you need to gather the information necessary to submit sales tax 2025?



EX 2.5 (LO 3) Data Managerial Accounting Apply Excel PivotTable Slicers Your supervisor at Best Bakes Bakery would like you to perform an analysis of costs by product. Create an Excel PivotTable that displays all products, and create slicers to provide the following information:

1. Total inventory costs for 2023.
2. Total inventory costs for cinnamon buns from 2022 to 2025.
3. Total inventory cost for cinnamon buns in 2023.
4. Total inventory costs for the city of Thornton from 2022 to 2025.
5. Total inventory costs for snickerdoodles in 2024 in the city of Brookfield.

EX 2.6 (LO 4) Data Managerial Accounting Descriptive Statistics You are a financial analyst at Animal Control Centers of America tasked with understanding the overtime pay for the year. Your IT group has provided a file that includes overtime pay per employee per month. The spreadsheet also includes the average monthly temperature. Perform a correlation analysis between the amount of overtime incurred and the temperature for each month.

1. What is the correlation coefficient for the amount and temperature variables?
2. Is the correlation between amount and temperature strong, moderate, or weak? Explain your answer.

EX 2.7 (LO 4) Data Managerial Accounting Descriptive Statistics The chief human development officer at Animal Control Centers has asked you to provide a summary report of overtime pay for the year. The IT group extracted data from the company's database and provided an Excel file. Provide a summary report by performing the following:

1. Generate the descriptive statistics for the data field Amount. Report the descriptive statistics output:
 - Mean
 - Standard Error
 - Median
 - Mode
 - Standard Deviation
 - Sample Variance
 - Kurtosis
 - Skewness
 - Range
 - Minimum
 - Maximum
 - Sum
 - Count
2. Create a scatterplot of overtime pay. The x-axis should be the month of the year, and the y-axis is the dollar amount of overtime paid.
3. Are there any unusual points on the scatterplot that need further investigation? Why or why not?

EX 2.8 (LO 2, 5) Data Tax Accounting Basic Excel Functions and Bar Chart Visualization You are a tax professional asked by the State of Wyoming's comptroller general to provide a report summarizing tax returns filed in the state. The information technology group has provided an Excel file that contains data regarding the zip code of filers, number of returns, and other important tax return data.

1. Which filing status had the most returns filed? (Single, Head of Household, or Married Filing Jointly)? Use an Excel function (not a PivotTable).
2. Create a column chart visualization showing the number of tax returns by filing status. The x-axis should include the filing status (single, MFI, Head of Household), and the y-axis should be the quantity of returns.

EX 2.9 (LO 2, 4, 5) Data Managerial Accounting PivotTables, Descriptive Statistics, and Visualizations You are a managerial accountant preparing an analysis of sales by segment. U.S. Outdoor Adventures has three segments: consumer, retail, and travel company. The consumer segment includes sales made to individual customers via the U.S. Outdoor Adventures website. The retail segment are those sales made to retail stores. The travel company segment are sales made to travel companies that organize and run camping trips.

1. Create an Excel PivotTable summarizing sales by segment from 2022 to 2025. Show each segment and the total sales by year.
2. Create a bar chart for sales by segment from 2022 to 2025. Which segments are increasing, and which segments are decreasing?
3. Create a PivotTable to analyze average sales by segment from 2022 to 2025. Show each segment and the average sales by year.
4. Create a line chart for average sales by segment from 2022 to 2025. Are average sales increasing or decreasing from 2024 to 2025?
5. **Dig Deeper:** Examine the variation in sales by segment using a PivotTable and line chart that displays standard deviation. Which segment has the most variation in sales? How did you determine your answer?

EX 2.10 (LO 2, 5) Data Auditing Pivot Tables and Line Graphs You are working on the U.S. Outdoor Adventure Company financial statement audit for the year ended December 31, 2023. Your senior has asked you to understand the sales data and identify significant customers on which to perform detail testing. The client has provided the sales data. According to the general ledger, total sales for the years ended 2025 and 2024 are \$273,323 and \$269,196, respectively.

1. Create an Excel PivotTable depicting sum of sales by year using the variable ShipDate. Verify the reported total sales for the year agrees with the client's general ledger balances as presented.
2. Create a PivotTable that presents sales for 2024 and 2025 by product category.
3. Using the best practices outlined in Illustration 2.57, create a line graph that shows sales for 2024 and 2025 by category. The x-axis should include the years 2024 and 2025, and the y-axis should have the dollar amount of sales. There should be three lines, one for each category of sales, camping gear, paddles, and tents.
4. **Dig Deeper:** Modify the line graph so the x-axis has quarterly sales information for 2024 compared to 2025 for each category of sales.

Problems

PR 2.1 (LO 3) Data Auditing Applying Filtering in PivotTables Your firm has been hired to perform an audit for Best Bakes Bakery. You must perform an analysis of profit by customer to determine if there are any unusual changes.

1. Create an Excel PivotTable that displays all customers and profit for 2024 in one column and 2025 in another column.
2. Use the Value Field Settings to show the percent difference from 2024.
3. How many customers have a percent changes in profit that is greater than $\pm 30\%$ from the prior year?

PR 2.2 (LO 2, 4) Data Auditing Excel Functions and Descriptive Statistics You are an auditor working on the financial statement audit of ThisBigCity asked to perform analytical procedures to understand the city's reimbursement expenses for the year ended December 31, 2025. The client provided a download of all the employee reimbursement data since 2010. To answer the following questions, use Excel functions, *not* PivotTables.

1. What is the total dollar amount of reimbursements paid in 2025?
2. Calculate the total dollar amount of reimbursements paid in 2025 to the following departments.

Department	Reimbursement paid in 2025
Department of Buildings	
Department of Health	
Department of Water Management	

3. What is the mean, median, and mode of the reimbursement amount in 2025?
4. What is the standard deviation of the reimbursement amount in 2025?
5. Generate a scatterplot depicting the reimbursement amounts in 2025. On the x-axis, show the date, and on the y-axis, show the amount. The y-axis range should be between \$-500 and \$3,500.
6. Use the following descriptive statistics and the scatterplot to identify any anomalies in the city's reimbursements during 2025.
 - Mean
 - Median
 - Mode
 - Standard deviation
7. **Dig Deeper:** Expand on this analyses to include a discussion of reimbursement amounts by department or by job title.

PR 2.3 (LO 2, 5) Data Auditing Basic Excel Functions and Pie Chart You are an auditor assigned to the Outdoor Adventure Company financial statement audit for the year ended December 31, 2025. Your senior wants to understand the sales data and identify significant customers on which to perform detail testing. Total sales for the year ended December 31, 2025, according to the company's general ledger, is \$273,323. (Note: sales are recognized when the product shipped to the customer.) Using the client-provided Excel file, perform the following:

1. Verify the data set is complete by summing the sales column and agreeing to the sales figure recorded in the client's general ledger. Write a sentence that indicates you have agreed the sales amount to the amount in the client's general ledger.
2. Using the best practices outlined in Illustration 2.57, create a pie chart that depicts sales by region for 2025 and identify the region with the largest sales. Is a pie chart the best visual for this question? Why or why not?
3. **Dig Deeper:** Analyze the data set to understand the company's largest customers. Present your analysis in the form of a visualization.

PR 2.4 (LO 2, 5) Data Financial Accounting Managerial Accounting PivotTables and Bar Charts You are an accountant for U.S. Outdoor Adventures preparing an analysis of sales for the monthly internal sales analysis report.

1. Create an Excel PivotTable to identify if there is a monthly sales pattern for total sales or from 2024 to 2025. Format the PivotTable so the sales amounts are in currency with zero decimal places.
2. Create a bar chart of monthly sales from 2024 to 2025. Does this chart help determine if there is a monthly sales pattern? Why or why not?
3. **Dig Deeper:** Create a PivotTable and a line chart to illustrate monthly sales patterns by product category in 2024 and 2025. What does the line chart reveal about product category sales patterns?

Professional Application Case: Pizza My Heart Food Truck, Inc.



In 2020 Sal Simonelli started Pizza My Heart Food Truck, Inc. with an old family recipe and one food truck in Fort Lauderdale, Florida. Pizza My Heart serves eleven types of pizzas, as well as breadsticks and chicken wings. They operate at local breweries in the Fort Lauderdale and Orlando areas. In 2021 Sal purchased a second food truck that his son, Franco, operates in Orlando, FL.

Sal and Franco believe their business is doing well and are considering expanding to other Florida cities. This is the income statement for the last two years.

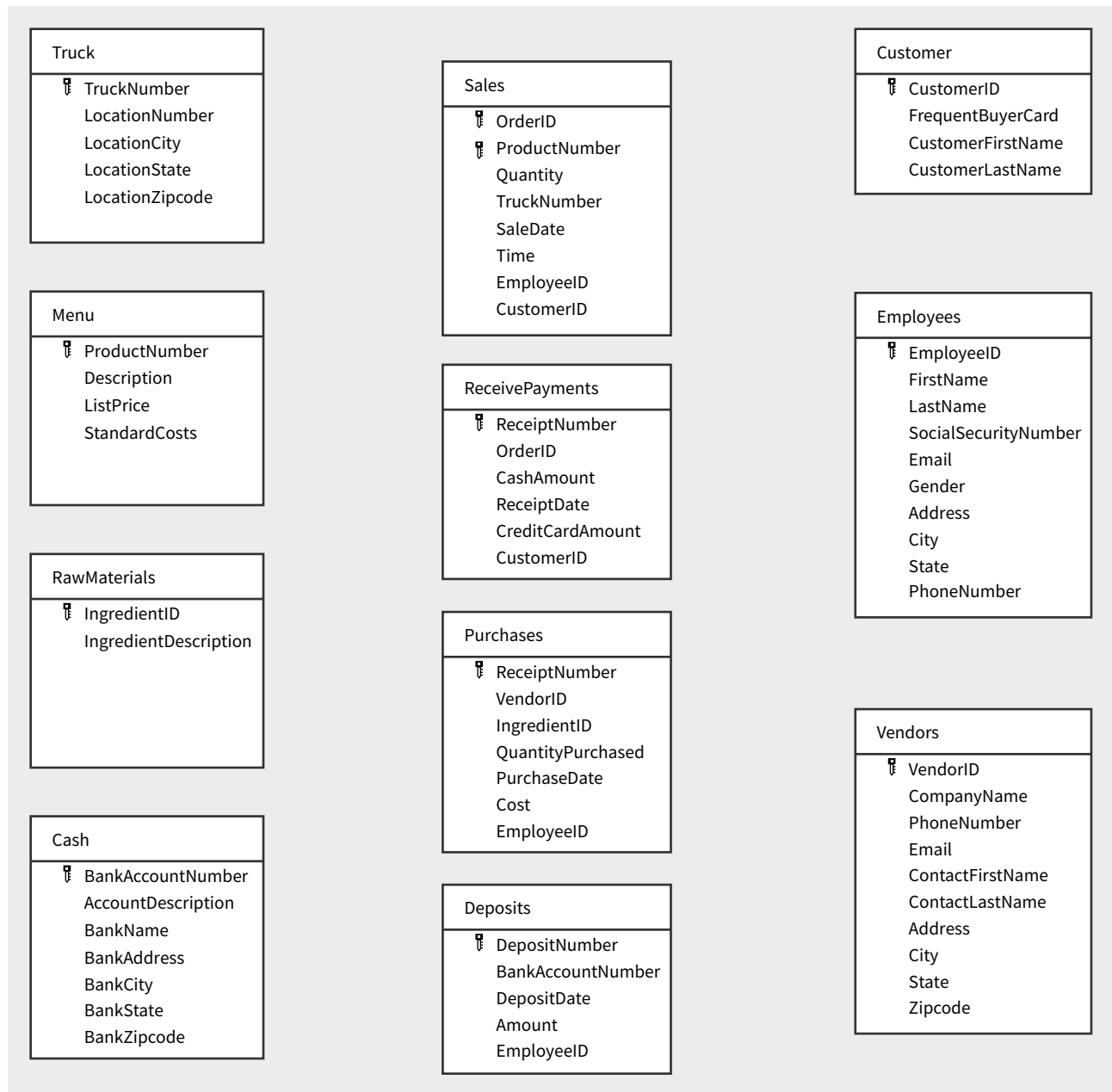
PIZZA MY HEART INCOME STATEMENT 2024 AND 2025

	2024	2025
Revenues		
Truck 1	\$151,200	\$333,396
Truck 2	\$100,800	\$222,264
Total Revenues	\$252,000	\$555,660
Expenses and Costs		
Cost of Goods Sold	\$138,600	\$305,613
Marketing	\$ 2,000	\$ 2,000
Salaries	\$133,890	\$155,600
Other Expenses	\$ 3,500	\$ 4,000
Total Expenses and Costs	\$277,990	\$467,213
EBITDA	(\$25,990)	\$ 88,447
Depreciation - Trucks	\$43,750	\$ 43,750
EBIT	(\$69,740)	\$ 44,697
Interest - Truck Loan	\$25,856	\$ 20,506
Pretax Income	(\$95,596)	\$ 24,191
Income Tax Expense	\$ 0	\$ 0
Net Income	(\$95,596)	\$ 24,191
Net Operating Loss Carryforward	(\$95,596)	(\$71,405)

- The business was operating at loss in 2024. This was largely due to the start-up costs of the second truck and because Sal and Franco were still searching for the best locations in Fort Lauderdale and Orlando for their trucks.
- The company began to turn a profit in 2025. Sales nearly doubled for the Fort Lauderdale truck and more than doubled for the Orlando truck.
- With the rapid increase in sales, Sal and Franco needed a better way to capture their financial and non-financial data.

They hired the accounting firm DGJ, LLC to create a database for them. The fields in each table are shown in the diagram. Primary keys are identified with a key symbol preceding the field name.

Pizza My Heart Database



PAC 2.1 Accounting Information Systems: Understand Relational Database Structure

Accounting Information Systems You are an accountant at the accounting firm, DGJ, that is helping Pizza My Heart create a relational database for their business. You have already created the tables shown previously. Now, you must create joins between the tables.

Identify the fields that would create a join between the following tables.

Tables	Fields
Sales and Customer	1.
Sales and Employee	2.
Sales and Truck	3.
Sales and Menu	4.
Employee and Purchases	5.
Deposits and Cash	6.
Purchases and RawMaterials	7.
Purchases and Vendors	8.

Consider each of the following questions, and list the tables and fields needed to answer them. Indicate whether the join should be a left, right, or inner join.

Question	Table	Fields	Join type
9. Are there vendors that have not made purchases?			
10. Do any employee addresses match vendor addresses?			
11. What is the total sales by menu item?			
12. What are the total sales by truck?			
13. Are there purchases made from vendors that are not in the Vendor table?			

PAC 2.2 Auditing: Create Analyses for Completeness of Cash

Data Auditing You are an auditor on the Pizza My Heart audit who was assigned the audit of cash. Use the cash receipts and bank deposit data files to complete the following analyses as part of your audit of cash for the first quarter. Use pivot tables where applicable.

1. Summarize daily sales, daily cash sales, and daily credit card sales by Truck 1.
2. Summarize daily deposits by Truck 1.
3. Prepare a reconciliation of daily receipts with daily deposits. Then, identify if there are any daily sales that do not match daily deposits.

PAC 2.3 Financial Accounting: Create Analyses of Sales

Data Financial Accounting You are preparing the first quarter sales results for Pizza My Heart.

1. Create a PivotTable that summarizes sales by Truck 1 and show the total sales for each month in the first quarter (January, February, March).
2. Create a PivotTable that summaries sales by product for the first quarter for Truck 1. Sort the data from highest sales to lowest sales.

PAC 2.4 Managerial Accounting: Create Analyses to Understand Costs per Product

Data Managerial Accounting You are a managerial accountant asked to prepare an analysis of costs. Use the purchases data file to answer the following questions with PivotTables:

1. What is the average of total costs per raw material ingredient? (Hint: use total costs, do not consider units).
2. What is the average cost per raw material ingredient by vendor? (Hint: Create two PivotTable calculations. Cost per unit of measure = Total Cost/(Size * Quantity) and Cost per unit = Total Cost/Total Quantity)
3. Are there vendors that are charging higher than average costs for ingredients?

PAC 2.5 Tax Accounting: Create Analyses to Plan for Expansion

Data **Tax Accounting** Pizza My Heart is considering purchasing another food truck, and Florida charges sales tax on food truck sales. Sal wants to choose the best location, not only for sales, but also for sales tax. Use the Florida population and sales tax by city data files to create a visualization that shows which cities have the largest populations and which also shows the tax rates by city.

1. What are the top ten most populated cities in Florida?
2. Use a PivotTable to determine the cities with the ten lowest tax rates.
3. Pizza My Heart is considering Tallahassee as a potential city for a new truck. Sal estimates that sales in the first year will be \$150,000. How much sales tax will be collected?

Le Grind Continuing Case: Use Descriptive Statistics to Analyze Sales Activity Over the First Three Years of Business

Data Visit Wiley's online learning platform for the case background, additional questions, data, and more details about the continuing case.

